Chapter 3

Derivative Product Examples

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Author: Ka Lok Chau

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3.1. Derivative warrants 窩輪

Derivative warrants refer to instruments traded in the stock exchanges. Theoretically speaking, the valuation is almost identical to options: Black-Scholes model is often used for pricing these instruments. There are two main kinds of warrants:

- warrants issued by a company on its own stock (sometimes known as *equity warrants*). New shares would be issued if warrants are exercised (known as a dilution effect). For example, in Hong Kong there is a warrant on APAC Resources (stock code 2478) (as of August 2025). This type of warrant is not too common, and they are rarely traded even though these instruments are listed in the exchange.
- the majority of warrants are issued by a party different from the underlying company, e.g. an investment bank (sometimes known as *derivative warrants*). The correct terminology is "covered warrant," although when people talk about "warrants" they often refer to this type. One example is a warrant on Tencent (stock code 20660) listed in Hong Kong, issued by the investment bank UBS. There is usually a conversion ratio (say 100 warrants control one share: the *conversion ratio* is 0.01, or *divisor* is 100). For example, if the divisor of a Tencent warrant is 100, 10,000 call warrants would entitle the holder to buy 100 shares of Tencent.

3.1.1. How do we compare different warrants?

Given that there are literally thousands of warrants listed in the market and it is common for a stock to have more than 20 or 30 warrants listed simultaneously which may have slightly different specifications (such as different strikes and maturity dates), we need some tools to compare them. The following benchmarks are often used:

$$\begin{aligned}
\text{gearing} &= \frac{S}{P \times div} \\
\text{premium} &= \frac{P \times div - (S-K)}{S} \times 100\% \text{ for call} \\
&= \frac{P \times div - (K-S)}{S} \times 100\% \text{ for put} \\
\text{Implied volatility: find } \sigma \text{ such that} & \\
P &= BS(S,K,r,d,t,\sigma)/div
\end{aligned}$$

In the above formulas, P is the warrant price; div is the warrant divisor; S is the spot price; K is the strike price; r is the risk free interest rate; d is the dividend yield; t is the time to maturity; σ is the underlying stock volatility; and $BS()$ refers to the Black-Scholes price.

As an example, we look at a European call warrant on Tencent (18902), issued by UBS. The warrant strike price is HKD 751.20, conversion ratio is 0.01 (or divisor = 100), i.e. 100 warrants entitle the holders to buy one share of Tencent. The maturity date is January 22, 2026. Minimum trading size is 10000 warrants, which is equal to 100 shares of Tencent (1 board lot). On August 7, 2025, Tencent's closing price was HKD 567.00, and the warrant price was HKD 0.101. Therefore, the parameters are:

Gearing (槓桿比率) = 567.00 / (0.101x100) = 56.139x

Premium (溢價) = {0.101x100 – (567.00 – 751.20)}/567.00 = 34.27%

Implied volatility = 35.854% (calculated from the Black-Scholes model with interest rate and dividend assumptions)

A gearing of 10x supposedly means that if you have $1, you are controlling an asset which is worth $10, and hence a 5% move in the underlying price would translate to a 50% profit or loss. However, from option theory, we know that when asset price moves up by $1, option (or warrant) price would move less (given by option delta).[1] For example, stock price = $20, warrant price = $2, conversion ratio = 1, delta = 0.4. When stock price goes up by $0.2 (=1%), warrant price would go up by 0.4x0.2 = 0.08 or 4%. The move is much less than what is predicted by the gearing, which is 10x (= 20 / 2).

A more representative benchmark is the effective gearing (實際槓桿比率), which is basically the gearing adjusted by option delta. In the above example, the effective gearing is 4. In the Tencent warrant above, delta calculated from the Black-Scholes model is 0.157, and thus the effective gearing is 56.139 x 0.157 = 8.74x. (For comparison, Tencent went down 0.264% and the warrant price went down 10.62% at the close on Aug 7, 2025).

3.1.2. Aspects of trading and hedging

Trading in derivative warrants provides another means to make money. However, we need to have a good understanding of the mechanics in order to benefit from the trading. There is a difference between the method in which an investor and a covered warrant issuer can make money from the operation. Investors are supposed to take a directional view of the market. For example, if stock price goes up (or down), normally the investor can make money by buying a call (or put) warrant. More sophisticated investors (such as hedge funds) can even try to trade based on the movements in volatility level. Some of the trading issues are discussed below.

On the other hand, covered warrant issuers seldom take directional views. In order to have a sustainable business, they will often try to sell the warrants at a higher price initially (compared to the fair value as obtained from the option pricing model). Afterwards, the traders need to dynamically hedge the position to reduce the fluctuation in profit / loss due to trading. At some point before maturity, the trader may even try to buy back the warrants at low "implied volatilities" in order to reduce the overall exposure. Furthermore, the inventory of outstanding warrants will be actively managed through market making activities (maintaining active buy/sell quotes, as required by the regulators).

Factors affecting option pricing

We mention above that buying a call warrant is one way how an investor may make money if the stock price moves up. But there are other market parameters which can affect the warrant price. In the example below, the option parameters are: strike price 100, interest rate 1%, maturity 6 months, no dividend.

	European Call Option			European Put Option		
Stock price	100	102	102	100	98	98
Volatility	40%	40%	35%	40%	40%	35%
Option price	11.469	12.623	11.221	10.971	11.873	10.495

[1] Delta is a hedging parameter, often defined as the change in option price when the underlying asset price changes by $1. Hedging parameters are discussed in chapter 8 of the course notes.

In the case of the European call, when stock price moves from $100 to $102 and the volatility remains unchanged, the option price has indeed gone up. However, when volatility moves from 40% to 35% at the same time, the option price has actually *decreased* compared to the initial level. The same occurred for the put. Say if an investor buys a European call when stock price is at $100, and hopes to make money when stock price moves up to $102, he/she will be surprised to find that the position suffers a loss. In these simple examples, it can be seen that even though the investor's view of the stock price movement may be right, the option price still moves in an opposite direction because of the change in volatility used. The implication is that the warrants may not give a "pure" exposure to the directional movement that the investor wants. Unfortunately, there is no way to prevent this unless the investor is ready to hedge the other risks.

Time to maturity, moneyness, and leverage

Option parameters: European call, strike price 100, interest rate 1%, no dividend.

Time to maturity	**6 months**				**1 week**			
Stock price	95	96	110	111	95	96	110	111
Option price	8.831	9.330	17.757	18.453	0.520	0.732	10.119	11.086
% move		5.65%		3.92%		40.77%		9.56%
Leverage		5.37x		4.31x		38.73x		10.51x

These examples show that when the maturity is longer, the option is not too sensitive to stock price movement, but the leverage effect is much higher when the maturity is short and the option is out-of-the-money. In the markets, investors who want to make big bets would trade warrants with very short time to maturity (known as 末日輪) because of the extreme leverage. (Also see the discussion in section 3.6).

Is warrant trading a zero sum game?

There is often a misunderstanding that if the investors are going to make money, the warrant issuer must be losing, and vice versa. Of course, this is only true if both parties are un-hedged! In reality, it is possible that both the warrant investor and the warrant issuer make money (so both are happy). We illustrate the case with a simple example:

- Cost of hedging the warrant (i.e. warrant fair price) = $10.
- Issuer sells the warrant at $14 to investor.
- At maturity, one warrant pays $30, and the investor makes ($30–$14=$16).
- Issuer makes ($14–$10=$4) (assume no extra hedging cost).

Effectively the warrant investor gives up some of the gain to the warrant issuer (theoretical gain is $20, but the investor only gets $16). One can say that this is not "fair," because if the view of the investor is right, then the investor should make the whole gain. However, from the point of view of the issuer, we can also say that the issuer is providing some kind of service to the investor, so that the "profit" made by the issuer is actually a "service charge." The level of profit can then be determined by market forces.

Typically, the issuer has a short position in the option (warrant), hence if underlying stock moves a lot (especially when spot price is around the strike), the issuer would likely lose (because of high volatility). In practice, the issuer would seldom leave the option unhedged (with respect to delta risk). For the volatility risks, vega hedging is achieved from purchasing OTC or listed options e.g. through issuing Equity Linked Notes (usually at different maturities/strikes), see section 3.3.). For uncommon underlyings such as customized baskets of stocks, only delta hedge can be carried out (not vega hedge) since it is often not possible for issuers to buy options on an identical basket.

3.2. Callable Bull/Bear Contracts 牛熊證

A different kind of warrant is the callable bull/bear contract, which is also known as turbo warrant. They first appeared and became popular in Germany in 2001. The Hong Kong Stock Exchange introduced these contracts in June 2006, and they have become very popular among local investors. Within a few years, the trading volume is comparable to that of derivative warrants (from Hong Kong Exchange's Fact Book 2022/Monthly Bulletin).

	Market Share (%)	
Year	CBBC	Derivative Warrants
2011	10.84	15.39
2012	11.56	12.41
2013	8.36	11.74
2014	7.24	12.03
2015	7.11	17.43
2016	8.42	16.75
2017	5.52	13.95
2018	6.99	14.70
2019	9.33	11.30
2020	7.29	7.06
2021	5.33	6.72
2022	6.32	6.42
2025 July	3.50	2.40

3.2.1. Product description

The payoff of a CBBC bull warrant is equivalent to the call option, i.e. it pays max($S-K$,0) at maturity. However, the special feature in CBBC is that the spot price S must stay above a certain barrier level B if it is to stay "alive"; if spot price hits B or falls below B before maturity, the warrant will expire. Usually B is only slightly above K, and the warrant will pay an amount close to ($B-K$) if the barrier is hit.

A typical CBBC has the following terms:

- Current spot price 21400, strike 21000, barrier 21100, maturity 3 months;
- If spot price always stays above the barrier 21100 before maturity, the warrant pays (S–21000) at maturity;
- If spot price hits 21100 or below before maturity, it pays (21100–21000) = 100.

In Hong Kong, there is usually a time lag of three business days between launch date and issue date of a CBBC. In many issues, the spot price is quite close to the barrier. Hence the leverage of the product can be quite high (because the option premium is small). However, given the time lag mentioned above, many CBBCs cannot be launched successfully because the market level may move below (for bull warrant) or above (for bear warrant) the barrier before the CBBC is issued.

Another feature in the CBBCs in Hong Kong is that there is a minimum issued price of $0.25 for each warrant (which means that there will be a varying scaling factor depending on the underlying option price). Furthermore, since the minimum trading spread is $0.001, subsequently this tick movement may represent a significant % movement in the CBBC's price. For example, if entitlement ratio is 10000 for a Hang Seng Index CBBC, one tick represents a movement of 10 index points, whereas the intrinsic value of the CBBC is often only a few hundred points (as seen in the example below).

CBBC examples

Issuer:	JP Morgan	
Underlying:	Hang Seng Index	
CBBC code:	63460	63496
CBBC type:	Bull	Bear
Launch date:	Aug 4, 2025	Aug 4, 2025
Listing date:	Aug 7, 2025	Aug 7, 2025
Maturity date:	July 28, 2028	March 30, 2028
Issue Price:	$0.25	$0.25
Strike:	24350	25600
Call Level:	24450	25500
Entitlement ratio:	10000	10000
Funding cost at launch:	0.2114	0.2066
Initial Funding rate:	2.91%	6.08%

Financing costs

The price of the CBBC includes a component called funding cost, which includes such charges as the funding requirement for holding the underlying asset, as well as an adjustment for the expected dividends before maturity (because the CBBC is based on price level, not total return). However, the initial funding cost (sometimes as high as 20% or above) bears little relationship to the real funding cost after launch, which can be a variable. In practice this is a cost which would be lost if the call level is reached. We can illustrate its impact through an example:

- Spot level 21283, strike 20700, call level 20800, entitlement ratio = 10000, bull CBBC (c.f. the price of CBBC code 69047, issued by Guotai Junan Securities, as of Feb 8, 2023)
- CBBC price = $0.066 (intrinsic value = (21283–20700)/10000 =$0.0583)

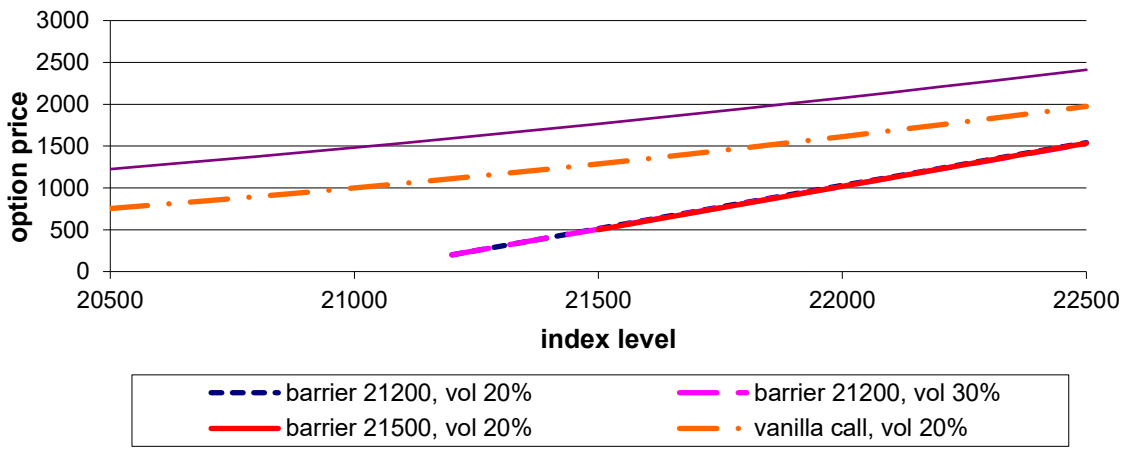
If spot index moves down to 20800 (i.e. 483 points move from 21283), assume that the redemption amount is only $0.010 (100 points), all the financing cost will be lost, i.e. 483 points move will lead to a loss of $0.066 – 0.010 = $0.056 (equivalent to 560 points). This should not be the case as, unlike the vanilla derivative warrants, theoretically the CBBC has no "time value" and any movement in the price level should accurately reflect the movement in the underlying spot price (see the discussion below).

3.2.2. Pricing and analysis

In option terms, the bull contract is a barrier call option, whereas the bear contract is a barrier put option. The standard barrier options may have an out-of-the-money barrier or in-the-money barrier, and the in-the-money barrier options are often very risky for both the buyer and the seller. The reason is because if the option can be knocked out while the option is in-the-money, the sudden change in the payoff can be caused by a very small movement in the underlying price, which means that the risk is very high. However, the interesting feature in the CBBC is that while the barrier is in-the-money, the option is also deep in-the-money initially. This alters the behavior of the barrier option completely, and the risks become more manageable.

One of the main reasons why CBBC is popular among investors is that, unlike vanilla options, this kind of barrier option is not supposed to be sensitive to volatility. Hence it is more transparent than derivative warrants. In section 3.1.3, we mention the case where the spot price moves up whereas the volatility moves down, and a long position in the call warrant would still lead to a loss. CBBC normally behaves much more predictably – if spot price moves up, the price of a bull CBBC will move up, irrespective of volatility movement. However, it is still possible that the movements in the CBBC's prices are affected by liquidity, thus it may not move exactly according to theoretical model prices. [In fact, many CBBC's were priced at a discount to the intrinsic value as of early October 2023, reflecting the market sentiment of the high volatility during this period.]

CBBC: bull contract pricing profile



The strike of these options is 21000. In these examples, it is assumed that the settlement price equals the barrier level minus the strike if it is redeemed early. The maturity of these options is 4 months, and the interest rate is 1%. In the case of a vanilla call, we can see that the option price changes a lot when volatility changes from 20% to 30% (e.g. at spot price 21500, the vanilla call price changes from roughly 1250 to 1750). However, we see

three (almost) overlapping lines for the barrier options, even when volatility changes by 10%. In fact, from the theoretical pricing model, the price of the barrier option is just the intrinsic value. For example, for the barrier 21200 call, at spot price 21500 the theoretical price is very close to 500, which is just (21500 – 21000). Changing the volatility has no impact to the price. The option gets knock out (i.e. its value becomes the intrinsic value) when the barrier level is hit.

CBBC price trend example

	HSI	Daily movement	CBBC	Daily movement	Equivalent index points
25/7	25388.35		0.051		
28/7	25562.13	+173.78	0.034	-0.017	+170
29/7	25524.45	-37.68	0.037	+0.003	-30
30/7	25176.93	-347.52	0.069	+0.032	-320
31/7	24773.33	-403.60	0.105	+0.036	-360
1/8	24507.81	-265.52	0.132	+0.027	-270
4/8	24733.45	+225.64	0.112	-0.020	+200
5/8	24902.53	+169.08	0.101	-0.011	+110
6/8	24910.63	+8.1	0.095	-0.006	+60
7/8	25081.63	+171.00	0.082	-0.013	+130

- Code: 59826
- Underlying: Hang Seng Index (HSI); Strike: 25830; Call level: 25730;
  Bull/Bear: Bear; Ratio: 10000.
  Launch date: 22/7/2025; Listing date: 25/7/2025; Maturity: 27/4/2028

The above table shows a real example of the price movement of a CBBC. During the life of this warrant, the actual daily movement roughly tracks the underlying index. Comparing the column of the daily movement of the HSI and the daily movement of the CBBC, at least the direction was always maintained. This is a considerable improvement compared to the derivative warrants.

CBBC: early redemption

When the barrier level is hit, the rebate is the difference between the settlement price and the strike. This barrier level can be triggered due to the intra-day movement of the underlying, and is not based on the closing price. In different countries, there can be different ways to determine the settlement price of the CBBC. For example, the turbo warrant introduced by Societe Generale in 2005 and the Hong Kong CBBCs (type R) derive

the settlement price from the minimum (for bull) or maximum (for bear) of the asset price up to the next few trading hours (or the next trading session). The Australian version is calculated by taking the average closing price of the asset for the next 5 business days after the barrier is triggered.

3.3. Yield Enhancement Notes (YENs) 高息票據

3.3.1. Yield enhancement features

As its name suggests, the return of Yield Enhancement Notes is enhanced compared to the vanilla interest rate products. Features that appear in many YENs include:

- Optionality

Investor sells an option to the issuer and receives the option premium, hoping that the option would be out-of-the-money.

- Barriers

The existence of barriers would reduce the probability that the specified condition will be satisfied.

- Path dependence

This is also a kind of barrier option. For example, a product may have a call option payoff (with strike = 100%) only if the stock price hits 90% of its initial level during the life of the option. There is a common illusion about the probability of occurrence of specific paths, and the investor would often not be able to work out the fair price.

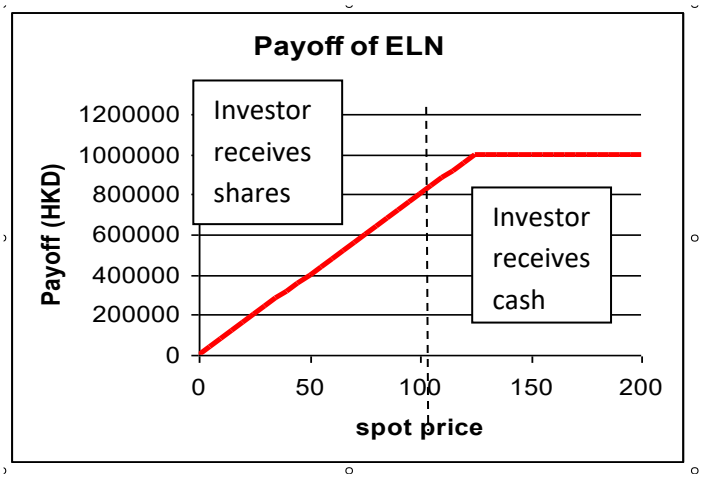
3.3.2. Basic YEN

YEN can be linked to different asset classes. The typical maturity is quite short, say 3 months or less. A popular type of YEN is the Equity Linked Note (ELN), with a stock as the underlying asset. The simplest ELN is one of the most popular structured products in the market. An example ELN may be:

Underlying:	Hang Seng Bank shares, strike = HKD 125
Maturity:	30 days (assumed 365 days in one year)
Notional amount:	HKD 1 Million

Equivalent number of shares = 1,000,000 / 125 = 8000

At maturity, if Hang Seng Bank is above HKD 125, investor receives HKD 1 Million; if Hang Seng Bank is at or below HKD 125, investor receives 8,000 shares (and some cash if it is an odd lot; not in this example).



Let's say that on trade date, the investor pays HKD 981,358 (98.14% of 1 Million) to buy the note. The equivalent yield r (deposit rate) if Hang Seng Bank stock price > HKD 125 at maturity is:

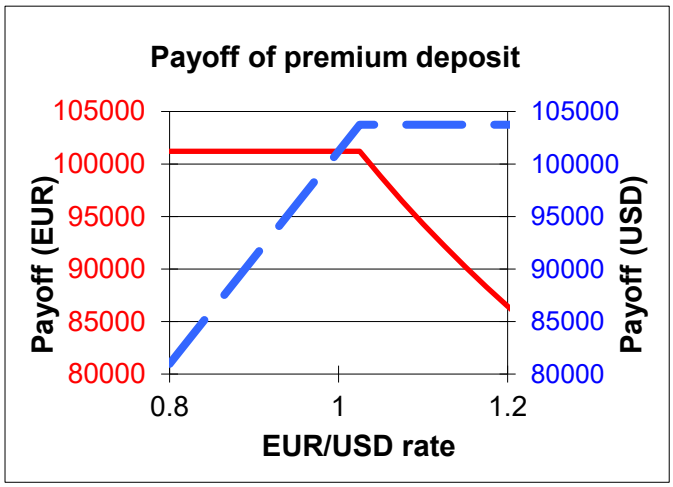
$$1000000/981358 = 1 + rt$$
$$= 1 + r \times 30/365$$

$$r = 23.11\%\text{p.a.}$$

The above diagram is similar to the profile of a short put position. Most ELNs can be broken into two components: *buy a YEN = buy a zero coupon bond (or place a deposit) + sell an option*. The option could be any type of option on any asset class. The most common example is that the option is *an at-the-money or out-of-the-money European put option on a stock* (sometimes known as a Bull Note or Bull ELN).

Another example of a YEN uses the FX rate as the reference underlying. One such product is known as a premium deposit. Typical terms are like the following:

- Spot FX rate: EUR 1 = USD 1.0100
- Base currency: EUR 100,000
- CHF/USD reference rate: 1.0250
- Tenor: 30 days
- At maturity, the bank has the right to pay: EUR 101,210.46, or USD 103,740.70 (=101,210.46x1.025)
- Equivalent deposit rate: 14.73% p.a., calculated by from 101210.46 = 100000 x (1+14.73% x 30/365)



In an FX contract, the payoff profile depends on the base currency selected. If USD is the base currency, the payoff profile is identical to that of the vanilla ELN. However, if EUR is the payoff currency, since the final amount has to be converted to EUR where the FX rate is in the denominator, the payoff profile (the red line) is not a straight line.

When should an investor buy YENs?

For YENs with embedded put options, investor is slightly bullish (i.e. hope that the stock price may not drop too far). The investor has a short put position, so the view must be that the stock would not be moving much. If stock price goes down, the option will be in-the-money and the investor can suffer a substantial loss. However, if stock price goes up, it would be better to buy stocks outright instead of earning the option premium only.

A common sales pitch would be something like this: choose a stock that is currently popular, where the investor is thinking of buying. Then the salesperson can tell the investor that he/she can potentially buy the shares at a discount (because the option strike is often set to be below the current share price). However, while the maximum gain is the enhanced yield, the potential loss could be big – the yield is only relevant if the option is not exercised!!! If the option has been exercised and shares were delivered, the investor has to be prepared to hold on to the shares as if they were bought at the strike price.

3.3.3. Pricing and early redemption

With the above example, we also have the following market parameters: spot stock price = $130, HKD 1-month interest rate = 3.85%, volatility = 30%, maturity 30 days, strike = $125.

- The amount that we need to deposit today to get $1 at maturity (i.e. price of zero coupon bond) = 1 / (1 + 0.0015x30/365) = 0.99988
- Price of 1-month put option on Hang Seng Bank, strike 125 = $2.315 = 1.852% of strike (from Black-Scholes formula). Note that this price is expressed as a % of the strike price, which is not the usual convention in other contexts as option price is more commonly expressed as a % of the current stock price.
- Price of ELN = 0.99988 – 0.01852 = 0.98136 or 98.136%

In the case of the premium deposit, we have the following input parameters:

- Spot FX rate = 1.01, EUR 30-day interest rate = 5.2%, USD 30-day interest rate = 1.2%, volatility = 14%.
- Price of 30-day call EUR/put USD option = 1.0785% (from an option pricing model)
- Bank charges EUR 300 at maturity (for example)
- Total amount received if option not exercised
  =100000x(1+1.0785%) x(1+5.20%x30/365) – 300
  = 101,210.46

YEN early redemption example

It is common for the investor to redeem the note early if enough profit has been made (usually within a short period of time). The close-out price just comes from a reverse pricing process, as illustrated with the market parameters given in the ELN example above.

- If stock price has moved up to $140 with remaining maturity 25 days, new option price from Black-Scholes formula = 0.276%; new note price is 99.99% (price of zero coupon bond) – 0.276% (option premium) = 99.714%.
- If stock price has dropped to $110 with remaining maturity 20 days, new note price is 99.99% (price of zero coupon bond) – 12.08% (option premium from Black-Scholes formula) = 87.91%.

The mark-to-market of ELN depends on both the interest rate and the option price, although the impact of interest rate is usually not significant because of the short maturity. Sometimes a commission fee is charged by note issuer for early termination, but it can also be incorporated into the calculation of option premium and a separate fee is not charged. In the above example, if the option premium does not include the commission rate and the rate is 0.20%, the investor will only receive 99.714% - 0.20% = 99.514% for early redemption.

3.4. Capital Guaranteed Notes (CGN)/Funds 保本票據／基金

3.4.1. Basic CGNs

One of the first types of structured products is the CGN, which was originated in the 1980's. It is also known as *principal guaranteed note or principal protected note.* For products with

equity underlyings, they usually have longer maturities than the yield enhancement products. Most products have maturities of at least one year, and sometimes can even be up to five years or more. For FX-linked products, maturity is usually limited to 3 to 6 months. The capital protection ranges from a minimum of around 90% to more than 100%, but the most common type is a 100% capital guarantee at maturity.

A typical CGN has the following terms:

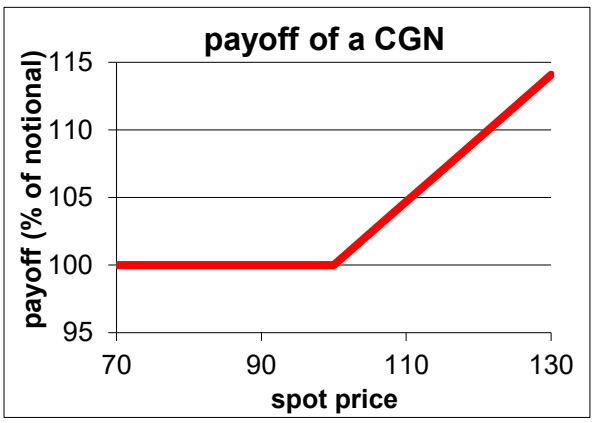
- Notional amount: HKD 10 Million
- Tenor: 3 years
- Guaranteed level: 100% of notional
- Payoff at maturity: 100% + P x R, where R is the return of the Hang Seng index, Max [(HSI_final – HSI_initial)/HSI_initial, 0]
  P is known as the **participation rate**, say 47%
- Usually the investor pays 100% of notional amount to buy this note initially.

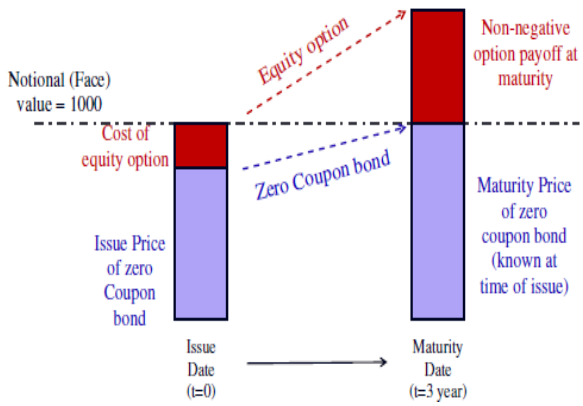
Another CGN example:

- Investor pays 100% of notional initially
- Tenor: 4 years
- At maturity, the investor gets the higher of:
  - 100%, or
  - 100% + P x A, where P is the participation,
  - A is the average of the monthly performance of a basket of S&P 500 & Eurostoxx 50 (positive performance only)

3.4.2. Pricing and early redemption

It can easily be seen that the CGN has the following payoff profile at maturity:





The profile is identical to that of a call option. In fact, *buying a CGN is equivalent to buying a zero coupon bond (or place a deposit), plus buying some options*. Theoretically the option could be any type of option on any asset class. The most common type of option is *an at-the-money European call option on an index or a basket* (which is also used in the second example above). We have seen many exotic options that can be embedded in these notes in the past. In some CGNs, *the structure is callable or can be early terminated*

based on particular market conditions; hence it cannot be broken down into simpler components (see the examples in section 3.5).

CGN calculation

We use the first example above and have the following parameters:

3-year interest rate r:	5.37%p.a.
Option premium:	24.546% (from option pricing model)
Bank's commission charge C:	3%

We would like to find out what is the participation rate for this note. Assume that the investor pays X = 100% to buy the note today, and would be guaranteed to get back G = 100% at maturity.

- Price of a zero coupon bond today $Z = G / (1+r)^T = 100\% / (1+5.37\%)^3$ = 85.48%. This is the present value of the guaranteed level G.
- Premium available to buy option = initial amount – PV of guaranteed level – commission = $X - Z - C$ = 100% – 85.48% – 3% = 11.52%
- Participation rate = number of options bought = 11.52%/24.546% = 46.93%

In fact, if we change the guaranteed level, we can come up with different participation rates. Higher participation is preferred as it should potentially give a higher return. A structurer can try to design a product which would be acceptable to investors based on the market sentiment at the time.

Guaranteed level N	Participation
90%	81.76%
95%	64.34%
100%	46.93%
105%	29.52%
110%	12.11%

Early redemption example

Given the long maturities of the CGNs, it is common that some investors may want to redeem the note early. In order to calculate the early redemption price, we can just make use of an identical pricing procedure. In the above example, assume that the remaining maturity is 2 years, spot price has dropped to 75 (used to calculate the option price), and 2-year interest rate is 4%. Let's say the option price is calculated to be 5.89%. Before taking into account of the commission cost, the note price can be calculated as:

$$\text{Price of a zero coupon bond} = 100\% / (1+4\%)^2 = 92.46\%$$
$$\text{Note price} = 92.46\% + 46.93\% \times 5.89\% = 95.22\%$$

Thus it is not guaranteed that the investor could get back the full amount if early termination is required, even if the guaranteed level is 100% or more. Actually, the guarantee only applies when the note is redeemed at maturity.

3.4.3. Additional remarks about the CGN

Usually, CGN will be more popular when the stock market is not performing well, because investors can be scared of further stock market losses. However, it does not work well in low interest rate environments because there will not be enough excess premium to buy an option, leading to very low participation rate which would not be too attractive for the investor. Therefore, we see the maturity of the notes/funds have extended from 3 years to more than 5 years, and moved from currencies such as USD or HKD to higher yielding currencies such as AUD. In recent years, vanilla CGNs become more popular again as interest rates for many major currencies have increased.

Finally, since the financial crisis in 2008, investors are becoming more aware of who is the issuer of the notes, due to the possible credit risk (i.e. probability of bankruptcy of the issuer).

3.5. Complex structured products 結構產品

Structured products are customized instruments that can provide specific exposures to various kinds of risks, with the potential of high returns if the favorable scenarios can be realized. Earlier we have already looked at some simple and popular products. In particular, a plain vanilla equity-linked note contains a simple put option, but the gain made by the investor is often limited because the maturity of these products is short.

The following list contains the features which would appear attractive to the investors:

- Ideally the potential absolute gain should be high, and not just the per annum return. For example, a product may return 18% p.a. but lasts one month, where the return is just 1.5%. On the other hand, a product that has a maturity of 6 months with 18% p.a. can generate a maximum return of 9%.
- On paper, the product should have potential of realizing a positive return for the investor, although the real probability of the event happening may be small.
- The underlying stocks for the product should be well known.
- Some popular products will limit the potential worst case loss, but the downside is that the potential positive return will be reduced.

Since the early 2000's, many structured products have been offered to investors. Investors buying these products would only want to assess the probability of gain, but they often have a naked position and won't hedge the risks. There is no need for them to know how the product is evaluated, as long as the payoff is known. As a result, when more and more product features are added, the complexity of valuation and hedging increase greatly.

Investment banks offering these products have to calculate the fair price, as well as designing a trading strategy to hedge the risks. As mentioned above, the products are complex and cannot be broken down into simple components. As the initial price of the product is often fixed at the nominal amount, the fair price would be embedded as part of the product feature, e.g. as the level of fixed coupon paid to the investor.

3.5.1. Example 1: SG Excel Notes

An example of a complex instrument: SG Excel Notes Series 37, was marketed by Societe Generale in Hong Kong in September 2008. It is not a capital guaranteed structure, and

the performance is linked to a basket of three underlying indices. Due to the financial crisis (Lehman Brothers' bankruptcy), the launch of the product was not successful. However, this kind of structure had been very popular in 2007/8, and was well accepted by investors even after the financial crisis.

Product description

- Initial price: 100% of the nominal amount
- Tenor: 1 year (minimum 1 month)
- Underlying basket: Tracker Fund (stock code 2800)
  A50 China Tracker Fund (stock code 2823)
  Hang Seng H-Share Index ETF (stock code 2828)
- Strike price: 89% of initial price for each fund
- Barrier price: 68% of initial price for each fund
- Valuation dates: every month from issue date to maturity date
- Callable price and schedule:

Valuation date (month)	1	2	3	4	5	6	7	8	9	10	11
Callable price (% of initial price)	100	99	98	97	96	95	94	93	92	91	90

- Bonus coupon: with respect to any valuation date other than the final valuation date, if the notes are redeemed early, a bonus coupon would be paid.

Valuation date (month)	1	2	3	4	5	6	7	8	9	10	11
Bonus coupon (% of principal amount)	1.4	2.8	4.2	5.6	7.0	8.4	9.8	11.2	12.6	14.0	15.4

- Early redemption condition: if the closing price of EACH underlying fund is at or above its callable price on a valuation date, the Note will be redeemed early at 100% of the principal amount.
- If notes are not redeemed early, there are two possibilities at maturity:
  - Noteholder will get 116.8% of principal amount if either:
    - Closing price of each underlying fund is at or above its strike price on the final valuation date; or
    - Closing price of each underlying fund on ANY date between issue date and final valuation date is at or above its barrier price.
  - Noteholder will receive shares in the worst performing fund, and the number of shares received is calculated by:
    - Number of shares = principal amount / strike price of the respective fund.

Payoff examples

Assume the initial prices are as follows:

	Initial Price	Strike Price	Barrier Price
Tracker Fund (2800)	$21.6	$19.22	$14.69
A50 China Tracker (2823)	$14.2	$12.64	$9.66
H-Share ETF (2828)	$123.8	$110.18	$84.18

Example 1: If the fund prices are:

Time from issue date	2800.HK	2823.HK	2828.HK	Price of each fund > callable price?
1 month	21.00	12.07	123.80	No
2 month	21.43	14.91	124.20	Yes

The fund will be redeemed after 2 months, and a bonus coupon of 2.80% will be paid, i.e. total payout = 102.80%.

Example 2:

Time	2800.HK	2823.HK	2828.HK
1	21.33	12.78	115.81
2	21.20	10.22	113.92
3	20.03	10.37	112.71
4	19.90	12.07	97.83
5	20.32	12.78	99.01
6	19.80	11.36	115.15
7	19.86	12.99	99.01
8	19.25	13.06	100.31
9	18.96	12.85	111.42
10	19.30	11.36	112.73
11	19.23	10.15	90.41
Maturity date	13.61	10.37	80.50

- Since not every fund's price is above its callable price in each month, the note is not redeemed early.
- If the closing price of each fund is greater than or equal to the barrier price on each day until the maturity date, then the final redemption amount is 116.80%.
- Otherwise, the fund with the worst performance is delivered, which is the Tracker Fund in this example.
- Number of units delivered = 10000 / 19.22 = 520.29 units per note, or 520 units + $3.95 (19.22 being the strike of the option).
- The loss in this example is 1 – 13.61 / 19.22 = 29.2%.

Pricing considerations

It is evident that the product cannot be broken down into simpler components as in the case of the standard ELN or CGN. We could make some brief comments in terms of how to price this product. Firstly, it is noted that we need to model the behavior of three stocks (indices). Therefore, a tree model / finite difference algorithm would not be appropriate as the dimension is too high. Secondly, the payoff profile is path dependent – there are early redemption conditions according to some pre-determined rule. It is a kind of barrier option with complex knock-out patterns. The inputs to the model should include the volatilities of each fund, correlation between each pair of funds, dividend assumptions, and the interest rate.

A Monte Carlo algorithm seems to be the only choice. However, we need to simulate the closing prices on each day in order to decide whether the barriers have ever been breached or not, which may reduce the efficiency of the algorithm. Furthermore, discretization errors can be high for barrier options. Special tricks may be needed in order to attain a correct evaluation.

For even more accurate modelling, we need to take into account of the correlation between the stock prices and the stochastic interest rates (used for discounting the cash flows). However, adding another level of complexity means that we have to define more input parameters, which may not be available from the market.

3.5.2. Example 2: HSBC's Basket DCDC ELI

A slightly simplified version of the above product has become very popular in recent years, which has been offered by various investment banks. It is known by a variety of different names, and one such product is the Basket DCDC ELI offered by HSBC. The full name of this product is "HKD Non-principal protected unlisted Daily Cash Dividend Callable Equity Linked Investments."

Product description

- Nominal amount: N = HKD 10,000
- Underlying stocks: HSBC (0005), CNOOC (0883), Petrochina (0857), Xiaomi (1810)
- Start date: February 14, 2023
- Expiry date: September 14, 2023
- Potential coupon: CP = 1.45% per month
- Initial spot price: closing price of each stock on the start date
- Strike price: K_i = 90% of the initial spot price
- Call price: C_i = 98% of the initial spot price
- Knock-in price: KI_i = 80% of the initial spot price
- Knock-in condition: at expiry
- Coupon dates: March 14, April 14, May 15, June 14, July 14, August 14, September 14

- From the start date until the expiry date, the investor will receive a monthly coupon of CP unless early termination occurs.
- Early termination condition [daily auto-call provision]
    - If on any date after the first coupon date, the closing price of the worst performing stock is higher than its respective call price C_i, the investor will receive 100% of the nominal amount plus an accrued coupon, equal to CP * num_days / total days, where num_days is the number of trading days between the call date and the coupon date immediately preceding the call date, and total_days is the number of trading days between the coupon dates immediately preceding and following the call date.
- If the ELI has not terminated early, the final payoff received by the investor is calculated as follows:
    - A fixed final coupon of 1.45%, plus one of the following:

- If the worst performing stock is above the respective knock-in price KI_i, 100% of the nominal amount (= N) [this is known as the expiry knock-in feature]; else
- Final payoff = N * Sp / Kp where Sp = closing price of the worst performing stock and Kp = respective strike of the worst performing stock (for cash settlement).

- In the above example, the knock-in condition can only be triggered at the expiry date. In another version of the product, a knock-in event can occur on any date after the start date. An alternative name of the knock-in protection is the *Airbag* feature.

Payoff examples

Assume that the initial prices are given as below:

	HSBC	CNOOC	Petrochina	Xiaomi
Initial spot price (100%)	$56.6	$11.4	$4.1	$12.84
Call price (98%)	$55.468	$11.172	$4.018	$12.583
Strike price (90%)	$50.94	$10.26	$3.69	$11.556
Knock-in price (80%)	$45.28	$9.12	$3.28	$10.272

The following scenarios show the potential payoffs received by the investor:

- *Case 1*: on June 2, the closing stock prices are: HSBC = 60.10, CNOOC = $12.42, Petrochina = $4.55, Xiaomi = $12.63.
- Earlier termination (i.e. auto-call) has occurred, and investor receives (% of N=$10000) :
  - 1.45% on March 14, April 14, May 15; plus
  - 1.45% * 13 / 21 + 100% = 100.8976% on June 2 (assume no settlement date lag), where 13 is the number of trading days between May 15 and June 2, 21 is the number of trading days between May 15 and June 14 (May 26 is a scheduled holiday).

- *Case 2*: on September 14, the closing stock prices are: HSBC = $58.50, CNOOC = $10.77, Petrochina = $3.99, Xiaomi = $10.44, and early termination has not occurred.
- Investor receives (% of N=$10000) :
  - 1.45% on March 14, April 14, May 15, June 14, July 14, August 14; plus
  - 1.45% + 100% on September 14.

- *Case 3*: on September 14, the closing stock prices are: HSBC = $57.20 (101.1% of spot), CNOOC = $8.21(72.0% of spot, worst performing), Petrochina = $3.05 (74.4% of spot), Xiaomi = $9.88 (76.9% of spot), and early termination has not occurred.
- Investor receives (% of N=$10000) :
  - 1.45% on March 14, April 14, May 15, June 14, July 14, August 14; plus
  - 1.45% + 100% * 8.21 / 10.26 = 81.469% on September 14; or
  - If physical settlement is chosen, 1.45% on September 14, plus 970 shares of CNOOC (bought at $10.26) + $47.8 (before transaction cost).

Analysis and pricing considerations

This product is simpler than the one given in the previous section. We can see that it contains many features as discussed in the introduction to section 3.5. From the investor's point of view:

- The underlying of the ELI are well known companies.
- Potential coupon of 7x1.45%=10.15% in 7 months (17.4% p.a.), with a minimum coupon of 1.45% in 1 month.
- There is a reasonable chance of getting back 100% of nominal (if the worst performing stock does not drop more than 20% in 7 months, or if the early termination condition (auto-call) is triggered).
- The knock-in level provides an additional level of protection for the investor as it enhances the possibility that the investor can receive 100% of the nominal amount. If a product specifies that the knock-in event can be triggered from the start date to expiry, the chance of this event occurring increases, which is worse for the investor. Hence the coupon for this product will be higher than a similar structure where the knock-in event can only occur at expiry, in order to compensate for the additional risk taken by the investor.

This product is also popular among issuers. As there is a reasonable chance of early termination, the potential of having a higher sales volume (and more commission received) would increase. However, the issuer needs to consider two issues in terms of pricing and hedging. Firstly, as this ELI is a complex product, the price does not come from a standard model, and a numerical scheme has to be used (a simplified procedure is given in the appendix). In the example above, 1.45% coupon per month is paid to the investor. But is this a fair coupon to pay? When sales commissions are taken into account, a likely pricing scheme would be:

- Using a trial and error method, find a combination of C_i , KI_i , K_i and CP such that the fair price of the ELI falls between 97% to 99% of nominal amount, taking into account of the margin required for the trading desk. This would involve a decision of the attractiveness of the product based on the coupon and the levels involved.
- The ELI is passed from the trading desk to the sales desk. It would be sold to the investor at a price of 100% of nominal amount. The difference between 100% and the fair price is the sales commission.

The trading desk would also need to consider the foreseen hedging difficulty, especially when it is difficult to hedge the correlation risks between the stocks within the ELI.

A quanto version of DCDC ELI

In some cases, products which appear to be similar to each other can possess additional pricing and hedging difficulties for issuers. It is very important to have a thorough understanding of the various product features and the risks involved. An introduction to the different forms and the pricing of these products are given at the end of this chapter. A typical example is a slightly more complex version of the ELI when the note currency is different from the currency of the underlying stocks. For example:

- Nominal amount: N = CNY 1,000,000
- Underlying stocks: Pingan Insurance (2318.HK), China Merchants Bank (3968.HK)
- Start date: January 20, 2023
- Expiry date: January 20, 2024
- Potential coupon: CP = 0.75% per month
- Initial spot price: S_1 (for Pingan Insurance), S_2 (for China Merchants Bank)
- Strike price: K_i = 89% of the initial spot price
- Call price: C_i = 102% of the initial spot price
- Knock-in price: KI_i = 70% of the initial spot price
- Knock-in condition: any date before maturity
- Coupon dates: monthly, from January 20, 2023
- If 100% nominal amount is not returned, the final payoff in shares= N * X / Kp where Kp = respective strike of the worst performing stock and X is the exchange rate between CNY and HKD on expiry date

There are two additional features compared to the example in the previous section. In this example, the knock-in condition is monitored continuously and the coupon should be higher (as discussed above). For the investment bank, the biggest difference in terms of hedging the product is the uncertainty in the number of shares in the final payoff. Unlike an ELI based in the same currency, the current product cannot be assumed to have a fixed number of options. The number of options depends on X, which is only fixed at expiry. This is known as a "quanto" feature, and the pricing algorithm involves a special adjustment term that takes into account of the volatility of the exchange rate and the correlation between the exchange rate and the stock prices. (Refer to the Appendix 3.A for pricing considerations.)

3.5.3. Example 3: Accumulators ("I kill you later") and Decumulators

This was a very popular private banking product starting in 2007 and 2008, especially with stock underlying. Accumulators based on FX rates are more common for institutional investors. Basically, it is a commitment to buy a fixed number of shares per day at a pre-agreed price (accumulator strike), which is often set to be below the current spot price. There is a possibility of early termination of the whole trade when certain conditions are satisfied. The typical maturity is 3 to 12 months, although short-dated accumulators are not so attractive to the investor in terms of the pricing.

Features of Accumulators

- Knocked out provision – the note will terminate early if the underlying asset price reaches a certain level, typically set to 5% above the current market price of the underlying asset. Both investors and investment banks like this idea. Because early termination would mean that the stock price has gone up, the investors will be making a quick profit (with possible high leverage, see below). For investment banks, the early redemption will free up the funds, and they can now persuade their client to buy another product within a short period of time, which means that there are more chances to earn sales commissions repeatedly.

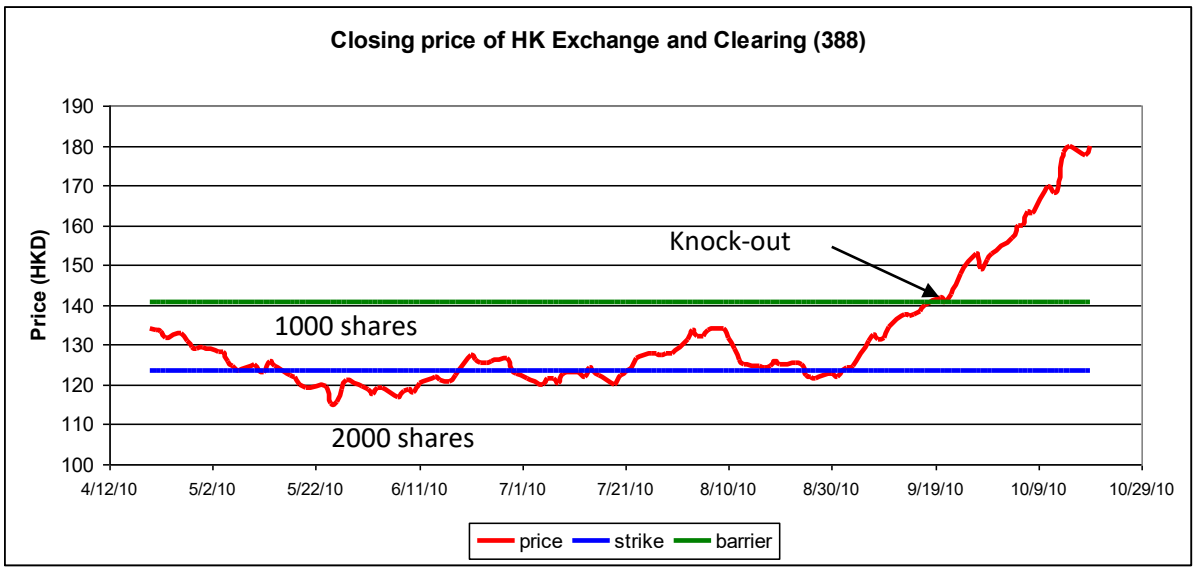
- Leveraged downside loss – if the market price drops below the accumulator strike, then the investor has to buy more shares. Leverage factor is often set to 2, but can be more during the period when accumulators were most popular. By including this feature, it is possible to design a structure with more attractive pricing levels, e.g. with a closer knock-out barrier, or a lower strike, which are both beneficial to the investor.

Accumulator example

Underlying:	Hong Kong Exchanges and Clearing Ltd (0388.HK)
Start date:	April 20, 2010
Maturity date:	October 20, 2010
Spot reference:	$134.00
Strike:	$123.30 (92% of spot)
Term:	Investor buys a fixed amount of shares everyday, unless a knock-out event occurs
Knock-out barrier:	$140.70 (105% of spot)
Knock-out event:	If the official closing price of the share on any trading day is greater than or equal to the knock out barrier, a knock-out event has occurred
Shares per day:	1000 or 2000 (up to a maximum of 252000 shares, assuming 126 accumulating days)

We can see that this note is actually a forward contract where shares will be accumulated irrespective of the share price. This is different from an option, where shares will only be delivered when the option is in-the-money. On each trading day prior to the occurrence of the knock-out event, 1000 shares will be bought at the strike price unless the official closing price of the share is lower than the strike price, when 2000 shares will be bought. The shares will be settled on a monthly basis. Thus, the maximum nominal amount = 252000 x 123.30 = $31,071,600, but the investment bank would normally finance part of the trade; for example, the bank would only require the investor to put 30% of this amount as margin.

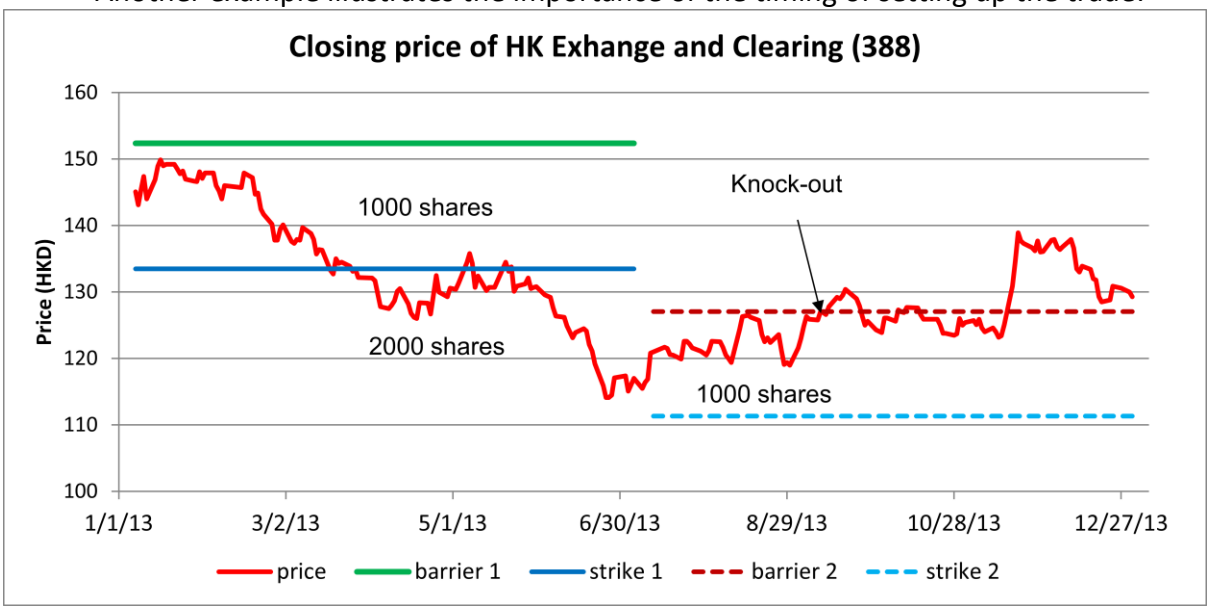
An example payout is given in the diagram below:



Total number of shares accumulated (assuming margin requirements are fulfilled) = 149000, while most days 1000 shares are accumulated per day, with a short period where 2000

shares would be bought each day. The above is a rare example where the investor can make a decent profit because of the substantial number of shares bought over a relatively long period of time (at $123.30) whereas the final share price rose above $140.70. Normally, extreme profit is not possible because of the knock-out feature – if the share price goes up initially, given that the barrier is only 105% of the initial share price, it means that the probability of knock-out is high and the number of shares bought will be limited. Alternatively, if the spot price falls below the strike, the investor can potentially suffer a substantial loss. Hopefully this would not happen in a generally rising market.

Another example illustrates the importance of the timing of setting up the trade:



Depending on the start date and the barrier, the performance can be very different. In the first example (a 6-month accumulator), if the start date is 1/7/13, the barrier is at $152.355, strike is $133.492, and the total accumulated shares in the 6-month period is 184000 shares @ $133.492 (final closing price = $117 on 7/5/13). A substantial loss will result. However, if the product is offered six months later and starts on 7/5/13, the barrier will be $127.05, strike is $111.32, and the total accumulated shares = 44000 shares @ $111.32 (with knock-out on 9/12/13; final closing price = $129.3 on 12/31/13).

Excursis: Citic Pacific's trades

Citic Pacific (0267.HK, part of the Citic group) is a Hong Kong registered company. According to the company's announcement on October 28, 2008, it has the following trades:

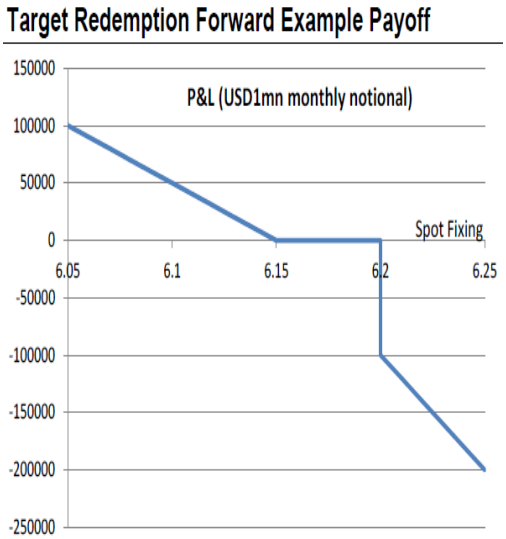
- Australian dollar (AUD) target redemption forward contracts, AUD 9.05 billion;
- Australian dollar daily accrual contracts, AUD 103.3 million;
- Dual currency (AUD/Euro) target redemption forward contracts (TRFs), AUD 290.7 million;
- Renminbi (RMB) target redemption forward contracts, RMB 10.4 billion.

These trades were supposed to be low risk trades, and provided some hedges to the operations of Citic Pacific in Australia as there may be substantial cash flows in the coming years. Upside potential for these trades was very limited.

Given that RMB was not freely convertible and was considered to have less fluctuation, the only surprising element in the above trades is the size of the AUD target

redemption forward contracts, which amount to AUD 9 billion. This is an enormous trade, which is much bigger in size compared to what was required in the Australian operation of the company. This contract is actually the currency version of the stock accumulator (see the section below). Citic Pacific was contracted to buy AUD and sell USD daily at an average strike of AUD 1 = USD 0.87 (the rate at the start of the contract was around 0.92). In fact, the daily trade amount was less than AUD 20 million, and the contract has a maximum term of three years. Maybe the company was led to believe that the trade would terminate well in advance (as was common for many accumulator trades) and did not realize its commitment when AUD FX rate dropped.

In any case, at AUD FX rate of 0.70, the loss of the company was HKD 14.4 billion. During the worst period, AUD dropped to around 0.60 on October 27, 2008. Citic Pacific announced the loss and was subsequently saved by its parent company, with both its chairman and CEO dismissed. It turned out that in 2009, the AUD FX rate went above 0.94, i.e. the initial knock-out barrier.



Target redemption forward is an FX version of accumulators, with knock-out level set to a target profit. As seen in the case of Citic Pacific's trades, this structure is popular with corporate customers. A typical trade can have the following terms:

- Maturity: 2 years
- Fixing every month: if the offshore Renminbi exchange rate (CNH) is below 6.15, sell USD 1 Million and buy CNH at 6.15; if CNH rate is above 6.20, sell USD 2 Million and buy CNH at 6.20
- Early termination: if accumulated positive profit is more than 4000 basis points (i.e. 0.4000 CNH per dollar), the structure terminates

In this example, the structure is initiated when the CNH rate is at 6.12. With CNH showing an almost constant appreciation trend, the product can be knocked out after a few months. According to one estimate, outstanding TRFs based on CNH reached a notional amount of about USD 40 Billion. When the exchange rate started to fall in mid-2014 and again in the summer of 2015, many institutional investors suffered substantial losses.

Pricing issues (Advanced Topic)

Let n = total number of observation dates
C_{uo} = up-and-out barrier call option
P_{uo} = up-and-out barrier put option
K and H are the strike and barrier
l is the leverage factor (say either 1 or 2)

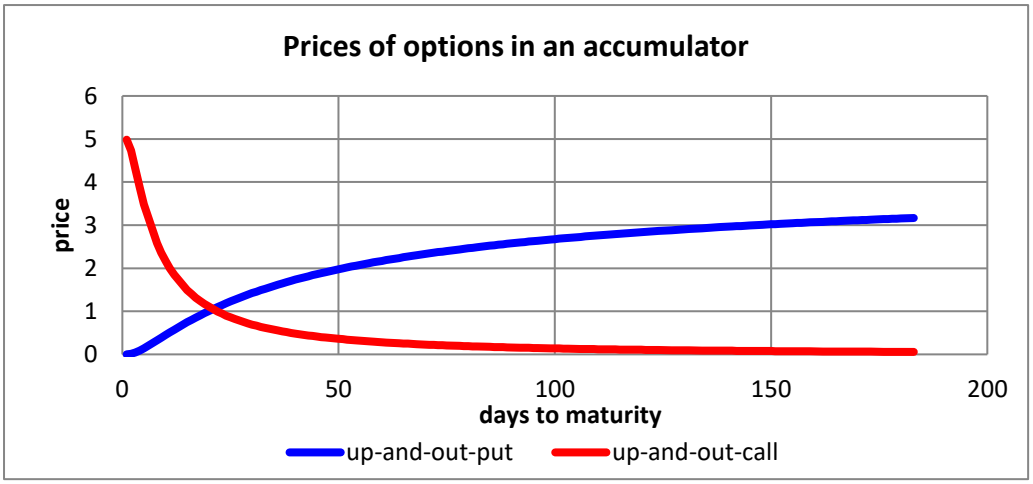
The accumulator is just a knock-out forward, and its fair price is given by:

$$F = \sum_{1}^{n} C_{uo}(t_i; K, H) - l \times P_{uo}(t_i; K, H)$$

The knock-out forward is priced as simultaneously buying an up-and-out call and selling (one or more) up-and-out put. Note that the settlement date of each option is usually delayed until the end of the accumulation period.

The reason why the accumulator product works is because of the interplay between the price of the call and the put. An example is given in the diagram below. We assume the following market parameters:

Spot price = 100, strike = 95, barrier = 105, volatility at strike = 32%, volatility at barrier = 30%, interest rate = 1%, maturity = 6months.

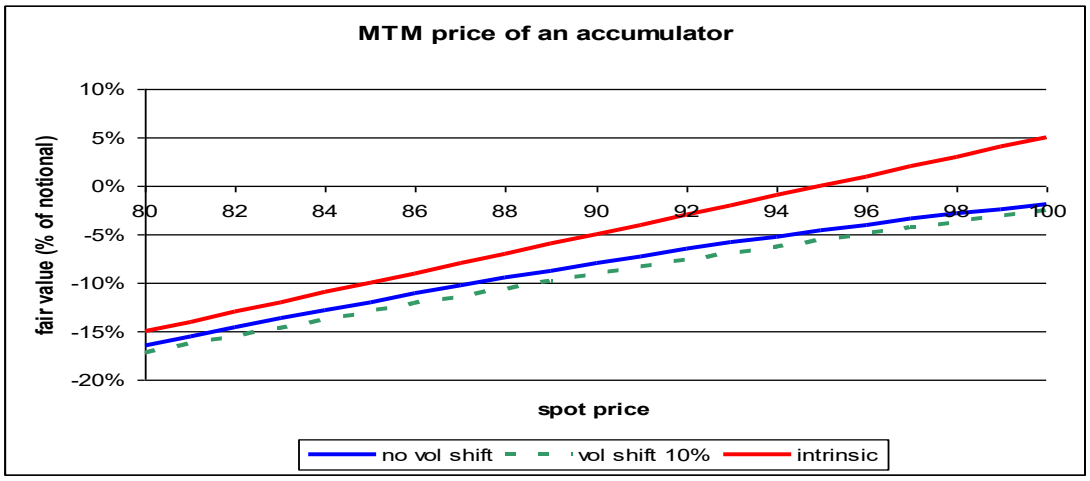


We note that the call options drop in value rapidly, whereas the put options are more valuable when the maturity dates are further from today. Given that the fair value of the accumulator is the difference between the two, if the maturity is short (say 50 days), the total of the call option prices would be much bigger than that of the total of the put options, which means that the accumulator's fair price is positive on trade date, and has to be sold at a price. However, all these structures are sold at no initial cost. In fact, it is usually structured so that the put options are worth more than the calls (i.e. the seller should have paid the investor, but pays zero in practice, and the difference would become the mark-to-market profit of the seller). Therefore, the maturity of an accumulator cannot be too short, because otherwise it is not possible to achieve a zero or negative cost.

Early redemption

When the market goes into a downturn, some investors may think that they want to cut the loss and terminate the contract early. What they do not often realize is that the put option

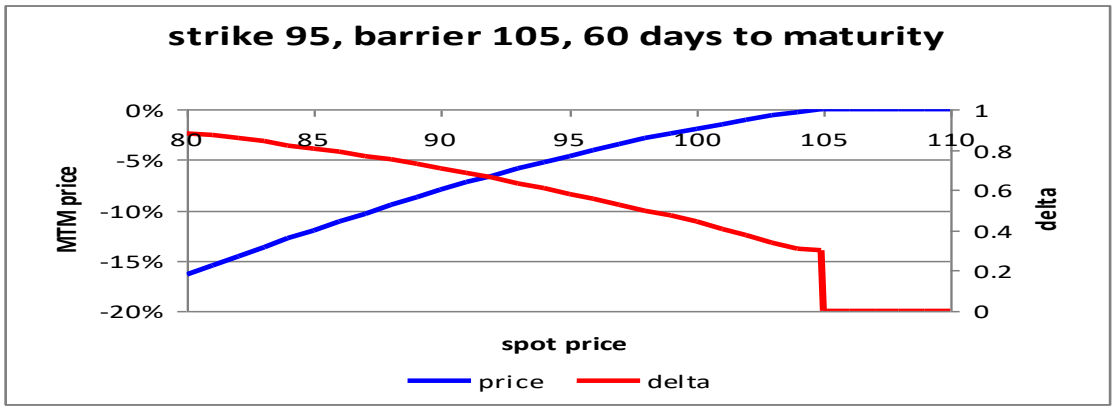
is very valuable, and the mark-to-market price of an accumulator is always more negative than the intrinsic value (although it is not too sensitive to volatility). In addition, the investor often needs to pay additional commissions to close out the contract early.



In this example, when the spot price is at 100, the intrinsic value is 5 (with the strike at 95), but the fair value is slightly negative (being the difference between all the calls and the puts). However, when spot price drops to 90 and if the investor wants to close out, the fair price is almost –8%, and this value is based on the total notional, not the daily trade amount. Hence the loss was often a surprise to many investors, and had been the subject of many private lawsuits during the financial crisis in 2008/09.

Risk profile of accumulators (Advanced Topic)

From the point of view of the issuer (investment bank) which would need to hedge the position, while an accumulator consists of up-and-out barrier calls (i.e. in-the-money barrier), the delta profile is quite smooth because of the aggregation, except at the barrier.



Accumulators: why have they been successful?

While the product was never popular in Europe or the US, accumulators have been successfully marketed to many private banking customers especially in Hong Kong and Singapore. In a trending and generally rising market, this product can lead to quick profit for the investor as the knock-out barrier is often set to be close to the current spot price. Many accumulators were based on blue chip stocks that were seen as safe investments.

A highly leverage return is achievable as it is not foreseen that the investor needs to use much capital. It is also a great product for the issuers as the quick turnaround time leads to repeated markup profit. Hedging the delta risk is not too difficult given the smooth profile. However, ultimately the investors suffered big losses when market dropped quickly; together with the leveraged returns, the losses were exacerbated.

The market for accumulators dropped to zero after the financial crisis in late 2008, but made a comeback when market sentiment improved. One estimate is that the amount outstanding in Hong Kong was about USD 5 billion in 2010 / 2011 (compared to a peak of about USD 23 billion in 2008). The investors learnt some lessons and the new accumulators have introduced a few risk mitigation features. Instead of a knock-out put, some of the current products would include a floor (i.e. like a put-spread) in order to limit the maximum loss. The maturity is shortened in order to reduce the total notional amount. There is no multiplier on the downside, and banks are less willing to provide a high leverage (i.e. they require more initial margin from the investor). While these measures would certainly reduce the attractiveness of the product because the pricing would be adversely affected, it would probably be used in a more prudent manner going forward.

Decumulators

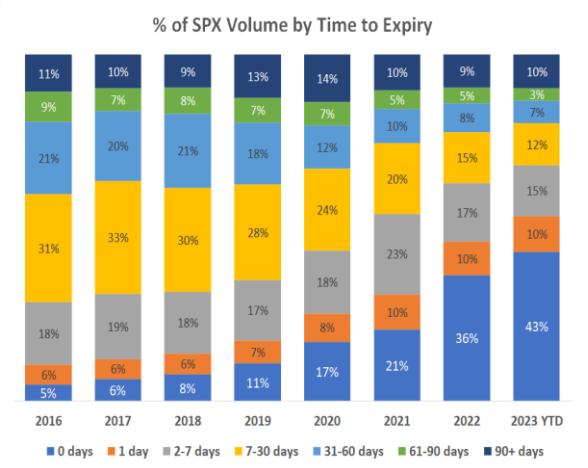
Instead of buying shares, a "put" version of accumulators, known as decumulators, has been introduced, which is sometimes used to help the investors of accumulators to dispose of their stock holdings. The risks can be analyzed as in the case of accumulators. A typical example is as follows:

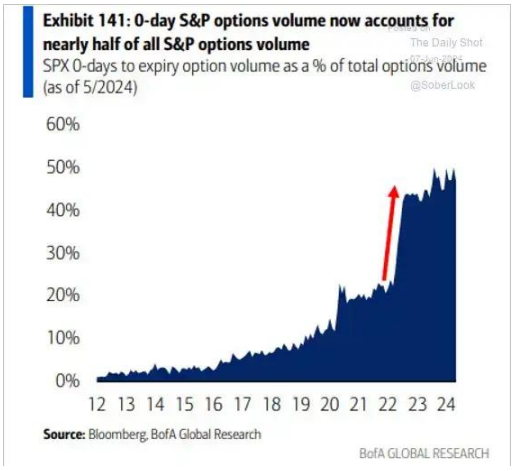
- Underlying: Petrochina (0857.HK)
- Start date: June 10, 2015
- Maturity date: June 10, 2016
- Spot reference: $8.82
- Strike: $9.88 (112% of spot)
- Term: Investor sells 10000 shares per day at the strike price, unless a knock-out event occurs
- Knock-out barrier: $8.38 (95% of spot)
- Knock-out event: If the official closing price of the share on any trading day is less than or equal to the knock out barrier, a knock-out event has occurred
- Settlement: Profit & Loss to be cash settled every month

3.6. Trading of 0DTE options

Traditional listed options have fixed settlement days, e.g. the third Friday of each month for the S&P500 options. The last trading time of these options is at the close of the previous trading day, and the settlement price of the contract is based on the opening level on the expiry date. Instead of trading these listed options, a popular strategy in the US market in recent years is the trading of the so-called 0DTE options (0DTE stands for 0 Day-to-Expiry). This has become possible when the CBOE started to introduce a series of S&P500 options

which expire using the closing price of the index (ticker symbols SPXW and XSP), and the settlement date occurs on every weekday. Trading of these options are allowed up to the final expiry time. At of August 2025, Hong Kong Exchange had announced plans to establish this type of option trading in Hong Kong by the first half of 2026.





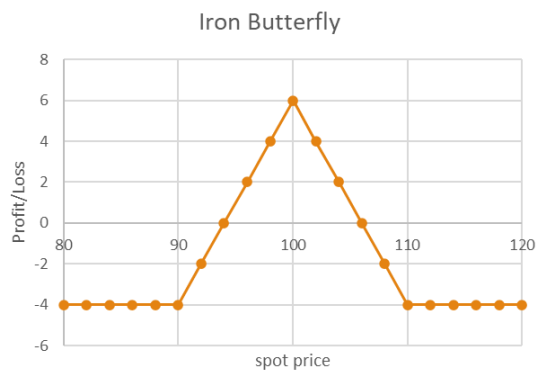
Source: https://www.cboe.com/insights/posts/volatility-insights-evaluating-the-market-impact-of-spx-0-dte-options/; BofA Global Research

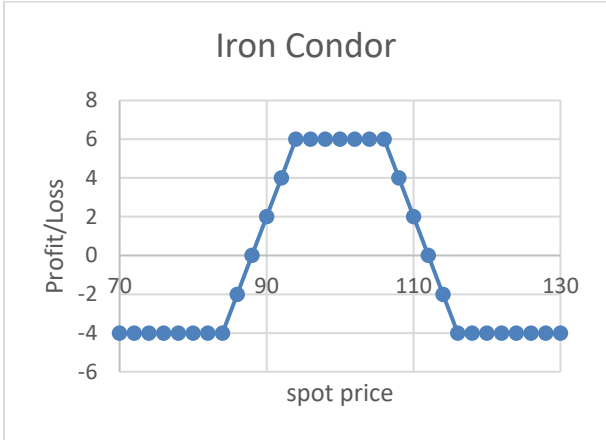
The above diagrams show the trading volume of listed options according to the time to expiry. As of May 2024, 0DTE options account for almost half of the trading volume of SPX options.

As discussed in section 3.1.3., options with short time to expiry is subject to high leverage, and small movements in the underlying spot price will cause significant movements in the option value. In the standard pricing models, volatility measures the variation between the returns based on the closing prices of the underlying asset. For options with less than one day to expiry, theoretical models (such as the Black-Scholes model) are less useful. Even if the time decay of the option is still calculated and we can measure time as a fraction and assuming the volatility accurately reflects the intra-day movement, the time value of the option is very small (due to the small t). The change in the value of the option mostly come from the change in the intrinsic value, which is caused by the movements of the price of the underling asset.

Common Trading Strategies

Retail investors are partly responsible for the high trading volumes of the 0DTE options. Two trading strategies are popular. These positions are often opened at the start of the trading day, and will be closed before the day close; sometimes the holding period can be as short as two hours.





A butterfly strategy can be created from options at three different strikes (c.f. section 1.3.3). In the case of 0DTE options, it is common to refer to the strategy as an iron butterfly. While different ways of construction are possible, a typical trade is the combination of shorting both a call spread and a put spread, with a common strike level K_2. For example, the call spread consists of shorting a call at strike K_2, and longing a call at strike K_3. The put spread is constructed with shorting a put at strike K_2, and longing a put at strike K_1. The strike levels K_1 and K_3 are often symmetrical around K_2, such that $K_1 < B_1 < K_2 < B_2 < K_3$. In the left diagram above, $K_1 = 90$, $B_1 = 94$, $K_2 = 100$, $B_2 = 106$, $K_3 = 110$. The diagram shows the total payoff including the initial option premium. A profit can be made if the spot price stays between B_1 and B_2. The strike levels are chosen such that the whole package is created as a net credit position, i.e. at the initiation of the trade, the investor is going to receive a (small) premium. The maximum profit is achieved if the underlying spot price has not moved from strike K_2. An important feature of the iron butterfly is the use of the out-of-the-money options as protection, so that the maximum loss is limited to the levels achieved when the spot price reaches K_1 and K_3.

A similar strategy is the iron condor. Instead of having a common strike for the call spread and put spread, the strike levels are separated. For example, the call spread consists of shorting a call at strike K_3, and longing a call at strike K_4. The put spread is constructed with shorting a put at strike K_2, and longing a put at strike K_1. In this set up, $K_1 < B_1 < K_2 < K_3 < B_2 < K_4$, and $K_1 = 84$, $B_1 = 88$, $K_2 = 94$, $K_3 = 106$, $B_2 = 112$, $K_4 = 116$ in the right diagram above. As in the case of the iron butterfly, a profit can be made if the spot price stays between B_1 and B_2, and the package is also a net credit trade.

These strategies would make money if the intra-day movement in the underlying spot price is not extreme. Taking into consideration of all the trading and execution costs, according to some accounts, in recent years the probability of making money from these trades is higher than 60% for the SPX options.

Appendix 3.A. Products involving two currencies

There are many occasions where domestic investors are interested in taking a position in a foreign asset. For example, a Japanese investor may want to express a market view on the S&P 500 index. However, any gains from the investment should be meaningful to the investor, and there are various ways to capture the exposure to the currency risk. Some products may be similar to one another, but there are often theoretical adjustments that need to be made to the standard pricing methods as discussed in Chapter 2.

3.A.1. Types of products

Typology
Four classical types of products can be created:

1. $C_1 = 1/X * \max[S' - K', 0]$. Foreign equity call option struck in foreign currency.
2. $C_2 = \max[S'/X - K, 0]$. Foreign equity call option struck in domestic currency.
3. $C_3 = 1/\bar{X} * \max[S' - K', 0]$. Fixed exchange rate foreign equity call option.
4. $C_4 = S' * \max[1/X - K_x,\ 0]$. Currency option with exposure on foreign equity.

S' is the stock price at maturity T, denominated in a foreign currency. K' is the strike price of the corresponding call option, denominated in a foreign currency. X is the exchange rate at maturity, expressed in terms of 1 unit of domestic currency = X units of foreign currency. K is the strike price of a call option, denominated in the domestic currency. $\bar{X}$ is a fixed exchange rate, as specified at the beginning of the contract. K_x is the strike price of a currency option, and its value should be comparable to X.

(2) is known as a composite option. (3) is also known as a quanto option. Of these four types, (1) and (3) are the most common. There is no problem with the analysis of (2), but it is not a popular product. (4) is rarely seen in the market.

Foreign equity call option struck in foreign currency

The domestic investor wants to express a market view on a foreign stock, but any gain is translated back to the domestic currency at the prevailing exchanging rate, and the investor is not concerned about the currency risk of the potential profit.

Example: USD European call option on N225 index
Payoff: USD $1/X$ x max(S' – 30000, 0)
where S' is the index level at maturity, X is the FX rate at maturity
Initial FX rate: USD 1 = JPY 140, or X = 140
(but this parameter is not relevant in subsequent calculations)

Note that if S' is 31000 and X = 145 (i.e. USD has strengthened against JPY), the final payoff is USD (1000/145). However, if S' is 29000 and X = 135 (i.e. USD has weakened against JPY), the final payoff is 0.

It is evident that if the option is out-of-the-money, we don't care what is the FX rate as there would be no payoff; hence the primary risk factor of the option is the volatility of the underlying asset. In fact, comparing this option with a JPY European call option at strike

30000, an investor sees no difference between the payoff of this option and the payoff of the equity call option in foreign currency, because any payoff is translated back to the domestic currency using the prevailing FX rate anyway. Thus the theoretical price of C_1 should not be different from the corresponding vanilla equity call option, expressed in the domestic currency at the prevailing exchange rate. No additional adjustment is necessary.

Foreign equity call option struck in domestic currency

Example: USD European call option on N225 index
Payoff: USD max(S'/X – 214.29, 0)
where S' is the index level at maturity, X is the FX rate at maturity
Initial levels: S = 30000, USD 1 = JPY 140, or X = 140,
at-the-money strike = 30000/140 = 214.29

If S' is 32000 and X = 145, the final payoff = 32000/145 – 214.29 = USD 6.40. However, if S' is 29000 and X = 135, even though the index level at maturity is lower than the initial level, the final payoff is 29000/135 – 214.29 = USD 0.52.

It is well known that the product of two lognormal variables is also a lognormal variable. S'/X can be treated as a new asset based on the domestic currency, which is USD in this example. The growth rate of this new asset is the domestic risk free rate (less the dividend yield of the foreign asset), and the volatility is given by the usual formula:

$$\sigma_{S'/X} = \sqrt{\sigma_{S'}^2 + \sigma_X^2 - 2\rho\sigma_{S'}\sigma_X}$$

The usual pricing framework can thus be used for the analysis of this product.

Fixed exchange rate foreign equity call option [Quanto]

The investor only wants to bet on the view of the underlying, but does not want to take FX risk.

Example: USD Quanto European call option on N225 index
Payoff: USD $1/140$ x max(S' – 30000, 0)
where S' is the index level at maturity
Initial FX rate: USD 1 = JPY 140, or X = 140
(but this parameter is not relevant in subsequent calculations)

Note that if S' is 31000 and X = 140, the final payoff is USD (1000/140). However, if S' is 29000, the final payoff is 0. The exchange rate at maturity does not affect the calculation.

Unlike C_1, we could not simply hedge this option with a JPY European call at strike 30000, because the payoff of this option may not match the payoff of the quanto option. Depending on the exchange rate at maturity, the payoff can be higher or lower than the quanto option. However, as in C_1, if the option is out-of-the-money, we don't care what is the FX rate as there would be no payoff; hence the primary risk factor of the option is still the volatility of the underlying asset. Hedging of this option will be discussed in Chapter 8.

3.A.2. Modelling quanto products[2]

Identifying the price process

Let

S = spot price of underlying asset in foreign currency

r_d, r_f = interest rates for the domestic and foreign currencies

q = dividend yield of the underlying asset

X = Exchange rate, expressed as 1 unit of domestic currency = X units of foreign currency

σ_s = volatility of the underlying asset S

σ_x = volatility of the exchange rate X

ρ = instantaneous correlation between the returns of S and X

Typically, we can model S and X as geometric Brownian motions, and the price dynamics are given as:

$$\frac{dS}{S} = \delta_S^f dt + \sigma_S dW_S^f$$
$$\frac{dX}{X} = \delta_X^d dt + \sigma_X dW_X^d$$

δ_S^f and δ_X^d are the drift rates of the variables, and define $dW_S^f dW_X^d = \rho dt$. The superscript f and d refer to the foreign and domestic risk neutral probability measures. As S is a stock denominated in the foreign currency, it's growth rate under the foreign risk neutral probability measure is the risk free rate r_f, adjusted by the dividend yield q:

$$\delta_S^f = r_f - q$$

For the pricing of quanto products, we need to identify the process for the foreign stock under the domestic risk neutral probability measure:

$$\frac{dS}{S} = \delta_S^d dt + \sigma_S dW_S^d$$

The critical term is the drift rate of this asset, as the volatility of the asset is not altered by the change of measure. The change of probability measure is given by the Girsanov theorem, and the drift rate is adjusted as:

$$\delta_S^d = r_f - q + \rho\sigma_s\sigma_x$$

As an example, we can use this formula to price a quanto forward. The forward price of the foreign asset at time T, denominated in the foreign currency is:

$$F_f = S_0 \exp[(r_f - q)T]$$

The forward price of the foreign asset at time T, denominated in the domestic currency (quanto forward), is:

$$F_q = S_0 \exp\left[(r_f - q)T\right] \exp(\rho\sigma_s\sigma_x T)$$

As a comparison, an asset denominated in the domestic currency with an initial price S_0 and a dividend yield q_1 would have a forward price:

$$F_d = S_0 \exp[(r_d - q_1)T]$$

Assume the following parameter values:

S_0=29000, r_f=1.5%, q=1%, ρ=0.3, σ_s=21%, σ_x=12%, T=0.25

[2] For a more precise treatment, see Kerry Back, *A Course in Derivative Securities* (New York: Springer, 2005), 116-123.

For the Nikkei 225 index denominated in JPY,

$$F_f = 29000\exp[(0.015 - 0.01)0.25] = 29036.27$$

For a contract traded in the CME, with the Nikkei 225 index denominated in USD,

$$F_q = 29000\exp[(0.015 - 0.01)0.25]\exp(0.3 \times 0.21 \times 0.12 \times 0.25) = 29091.20$$

In this calculation, the USD interest rate is not required.

For the pricing of quanto products, we can make use of the asset dynamics obtained above. Alternatively, we may treat "*S* denominated in the domestic currency" as a new asset, and make use of existing results. For example, we could use the Black-Scholes result for the pricing of a quanto European call option. Note that any asset denominated in the domestic currency with a dividend yield q_1 will have a drift rate of $r_d - q_1$. The price of a European call option on this asset is given as:

$$C = S\exp(-q_1 t)N(d_1) - K\exp(-r_d t)\,N(d_2)$$

The price of the quanto call option can be found by equating the drift rate:

$$r_d - q_1 = r_f - q + \rho\sigma_s\sigma_x, \qquad or \qquad q_1 = r_d - r_f + q - \rho\sigma_s\sigma_x$$

The new drift rate is incorporated into a new "dividend" rate of q_1, and the existing Black-Scholes formula could be used to obtain the price of the quanto option.

The meaning of ρ

The above expressions assume the convention that the exchange rate is expressed as 1 unit of the domestic currency = X units of the foreign currency, and the correlation between S and X is defined accordingly. However, exchange rates are governed by market conventions, e.g.

USD 1 = JPY 142.15
GBP 1 = USD 1.2795
CNY 1 = HKD 1.0750
USD 1 = CNY 1.1322

If the correlation parameter is obtained from an analysis of historical data, it is important that the sign of the parameter refers to the definition above.

Example 1: Nikkei 225 Index denominated in USD

- USD is the domestic currency, JPY is the foreign currency
- the calculations already refer to the correct convention.

Example 2: S&P500 Index denominated in GBP

- GBP is the domestic currency, USD is the foreign currency
- this situation is the same as example 1.

Example 3: Hang Seng Index denominated in CNY

- CNY is the domestic currency, HKD is the foreign currency
- this situation is the same as example 1.

Example 4: S&P500 Index denominated in CNY

- CNY is the domestic currency, USD is the foreign currency
- However, the usual convention for the exchange rate for this currency pair is not expressed as CNY 1 = USD P. If we calculate the historical correlation between S&P and CNY and this parameter is *0.25*, we need to input a value of *−0.25* in the quanto calculations. In general, if the correlation between two lognormal variables, such as the returns of A and B, is r, the correlation between the returns of A and $1/B$ is $-r$.

Appendix 3.B A simplified procedure for pricing DCDC ELI

The following procedure applies to DCDC ELI where the knock-in condition can be triggered at any date after the start date, and the auto-call condition is valid after the first coupon date.

1. Use the Monte Carlo method to generate a price path; suggest to use about 30 time steps for each month between the start date and expiry date. For example, if the expiry date of the product is 6 months, use 180 time steps.
2. From the first step, check whether a knock-in event has occurred. Record the event once this condition is satisfied. This check is performed at each step before expiry unless a knock-in event has already occurred or the product is terminated early.
3. On the first coupon date, apply the first coupon. Calculate its present value to the start date and record this value. Add this to the total payoff.
4. From the first call date onward (=first coupon date), check whether the auto-call condition has been reached. If yes, the product is terminated. Record the total coupon paid to the investor, apply the present value and add to the total payoff. Go to step 8.
5. The price path is extended by one day, and step 4 is repeated until the next coupon date.
6. Apply the next coupon. Calculate its present value to the start date and accumulate this value to the total payoff. The path is extended by one day, and repeat steps 4 and 5 until the expiry date is reached (remember to apply the coupon on the expiry date).
7. On expiry date (only reach this step if the product has not terminated early), determine the payoff taking into account whether a knock-in event has occurred. Calculate its present value and accumulate to the total payoff.
8. Repeat steps 1 to 7 with R Monte Carlo runs. The average payoff is the fair price that should be paid by the investor, based on a coupon of CP%.
9. The profit to the investment bank is given as: Price sold to investor minus the fair price of the product.

Note:
The procedure described above is only a first approximation. Financial companies would make use of more complex models with some of the following features:

i) Pricing any structure with knock-in and auto-call features by the Monte Carlo method is not very accurate due to the discretization errors. Special tricks need to be applied to smooth the knock-in and auto-call conditions. Also, details of business days and any potential dividend needs to be accounted for.
ii) This structure is sensitive to the distribution of the underlying stock prices. A lognormal distribution with constant volatility is perhaps too simple. More sophisticated volatility models are often used.
iii) For structures with longer time to expiry, it may be necessary to take into account the correlation between the underlying stock price and the discount rate used to calculate the present value (i.e. assuming both the stock price and interest rates to be stochastic).

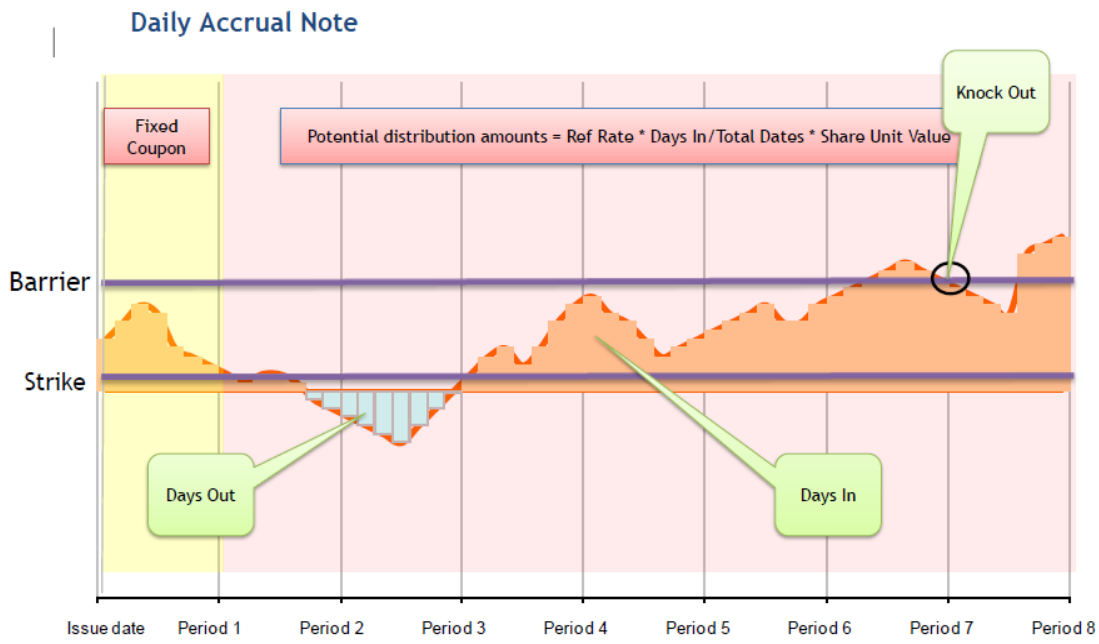
Appendix 3.C. Further examples of complex structured products [Advanced Topic]

The popularity of complex structured products reached its peak before the financial crisis of 2008. In this section, we review some of the exotic structures of the past, and discuss the theoretical and practical challenges. Some of these ideas can still be applied to the market conditions today.

3.C.1. Equity accrual note with barrier and path dependency

We give an example of an ELN which has a path dependent payoff.

Maturity:	2 years
Underlying stock:	Sun Hung Kai Properties
Reference stock price:	$120
Strike price:	$110.4 (92% of reference stock price)
Barrier price:	$130
Coupon:	2% for the first 3 months 2% x n / N for every 3-month period if there is no early termination n is the number of days that stock price > strike
Early termination:	if the closing price of the stock is higher or equal to the barrier at the end of each 3-month period, the issuer has to redeem the note at 100%
Final redemption:	100% of par if closing price is higher than $110.4; or physical delivery of [notional amount/strike price] in shares if closing price is below $110.4



Source: Hans Buehler, www.quantitative-research.de

Unlike the simple ELN, this note cannot be broken down into a summation of a zero coupon bond plus an option. The whole product has to be priced together, as there is a possibility that the note can be redeemed early, which is triggered by a condition based on the stock price. For the investor, the maximum payout of the note is limited to the coupon (8% per year), but the downside risk is similar to a short position of a put. Dependent on the stock chosen, a main selling point of this kind of structure is that the product is considered to have a high chance of getting a positive payoff (from the coupon payout) if the stock price is not too volatile, and there is also a possibility of early redemption (which is good from the investment bank's point of view in terms of generating business.

Another example of an exotic path dependent ELN is from a retail product: BOCI Flexibond - bonus ELN.

Maturity:	2 years
Underlying:	based on a basket of six Hong Kong stocks
Coupon:	10%
Coupon payment date:	6 months from trade date
Early termination:	The note will terminate if any of the condition below is satisfied

Time	condition (% of initial price)	Redemption price
1 yr	all stocks > 96%	110%
1.5 yr	all stocks > 92%	115%
2 yrs	all stocks > 88%	120%

Final redemption:	If the note is not redeemed early, at maturity, shares of the worst performer within the basket will be delivered, with a strike of 88%.

The pricing of this note is complex. If we ignore the early termination condition, price of the ELN = price of zero coupon bond + PV of coupon – price of a worst-of option, where the worst-of option: payoff = max (88% – S/S_0, 0), where S is the worst performing stock in the basket and S_0 is the initial price of that stock.

Note that the price of the worst-of option is strongly dependent on the correlation of the stocks in the basket (see the discussion in section 2). Without access of a pricing model with modeling assumptions and the accessibility of relevant market parameters, the investors could not possibly know what is the fair price of this note (e.g. is paying a 10% coupon fair? Should it be 12%? Or 9%?). Similarly, the salesperson of structurer can only treat the model as a black box when the product is put together (e.g. by choosing the stocks in the basket, or deciding how many stocks should go into the basket). Therefore, while this kind of product can have certainly intuitive appeal, the accuracy of the pricing model plays an important part in determining the product terms.

3.C.2. Products related to correlation among assets

3.C.2.1. General characteristics

These exotic structures are mostly seen with equity underlyings. They are often tied with thematic baskets, which can be based on recommendations from the investment bank's equity research department. Intuitively some of these products are appealing as the pricing can be much improved on the vanilla products. In general, investors will not be able to work out the probabilities when a payoff will result, as concepts of correlation are difficult to grasp.

The pricing algorithm is usually not a big issue. A lot of these products are strongly path dependent, and Monte Carlo simulation is the universal method. However, model risk can be high though, and how to hedge the risks is the most significant problem in these products. In particular, correlation and skew risks are difficult to manage because of lack of market instruments.

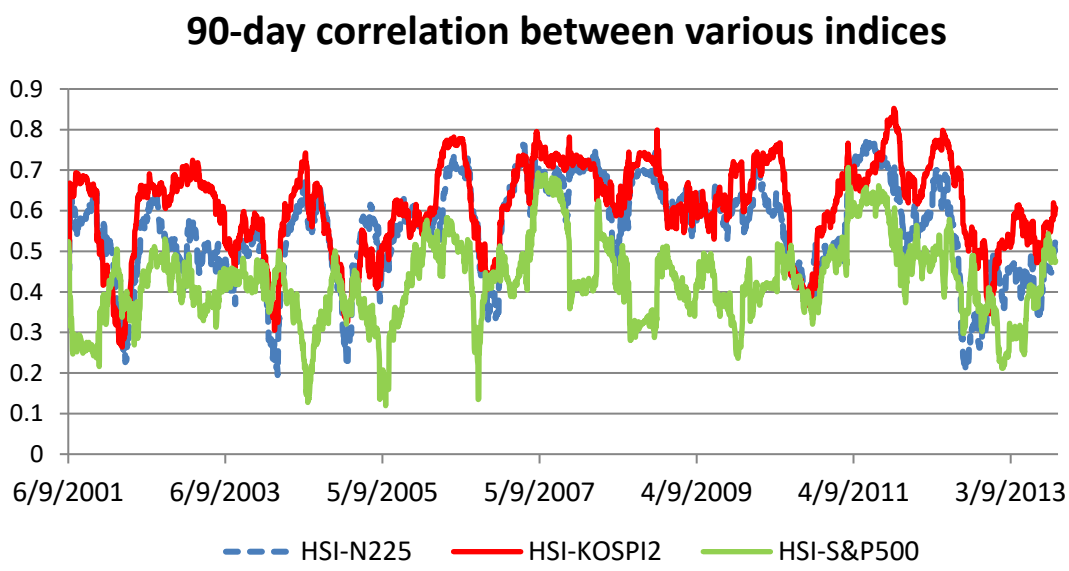
3.C.2.2. Understanding correlation

The formula used in calculating correlation is standard. The quoted figure is usually calculated as the correlation between the log returns of the underlyings:

$$\rho_{xy} = \frac{\sum_{n=1}^{N}\left(x_n - \bar{x}\right)\left(y_n - \bar{y}\right)}{\sqrt{\sum_{n=1}^{N}\left(x_n - \bar{x}\right)^2 \sum_{n=1}^{N}\left(y_n - \bar{y}\right)^2}}$$

$$x_n = \ln\frac{S_{1,n+1}}{S_{1,n}}, y_n = \ln\frac{S_{2,n+1}}{S_{2,n}}, \bar{x} = \frac{\sum x_n}{N}, \bar{y} = \frac{\sum y_n}{N}$$

However, the historical correlation between two underlyings can be quite unstable. The following diagram shows the typical short term correlation between pairs of equity indices:

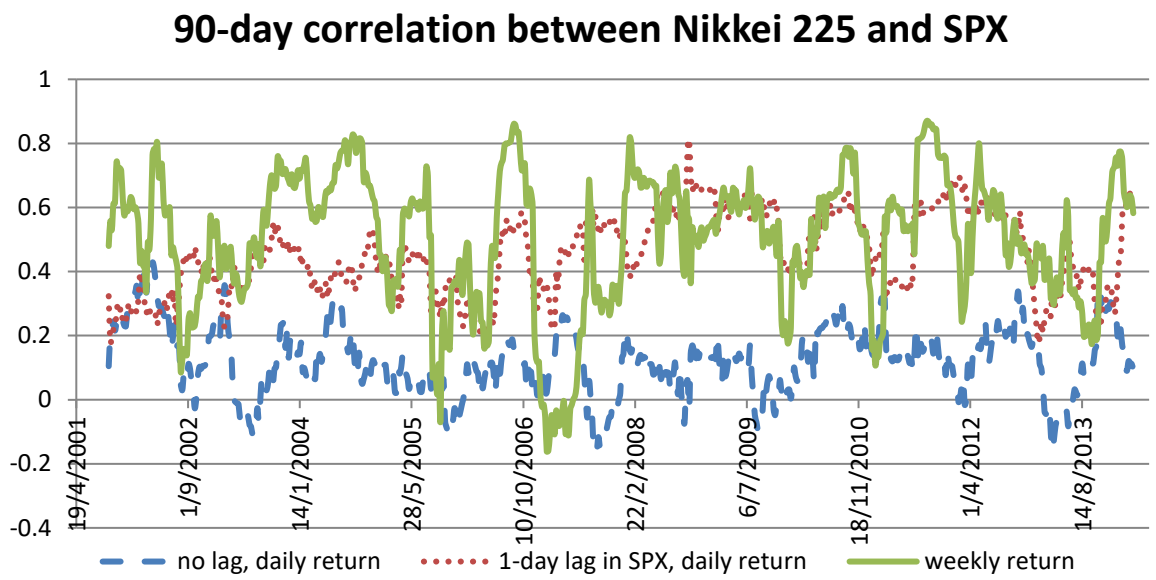


A commonly noted phenomenon is that during crisis periods, arguably the correlations between different stocks (or indices) will become higher.

There is also the issue of indices from different time zones. In particular, the US market closing time is often behind the other major markets. However, since incidents in the US market can have worldwide effects, it is common to try to take into account this development. Two methods are often employed:

- Measure the correlation with a 1-day time lag in the closing date. The HSI/S&P500 line above is calculated in this manner.
- Measure the correlation between weekly returns instead of daily returns.

The graph below shows the differences between these methods. It would not be easy to say which method is "correct," and one has to apply certain judgment in making the decision.



Implied correlation

With more than two assets, the pricing models normally require a full correlation matrix between all pairs of assets. Maintaining this correlation matrix can be difficult if there are many underlyings (e.g. some structures can have 20 assets, which means that there are 190 correlation parameters alone). Sometimes an approximation is used, by usng just one single "implied" correlation, assuming all correlations are the same. With this method, we can use the market ATM implied volatilities for the undelryings instead of historical volatilities.

$$\sigma_{index}^2 = \sum_{i=1}^{n} w_i^2 \sigma_i^2 + 2 \sum_{1 \le i < j \le n} w_i w_j \sigma_i \sigma_j \rho_{ij}$$

$$= \sum_{i=1}^{n} w_i^2 \sigma_{i,implied}^2 + 2 \sum_{1 \le i < j \le n} w_i w_j \sigma_{i,implied} \sigma_{j,implied} \rho_{implied}$$

The above formula gives the implied index correlation, but the same formula can be applied to calculate the historical realized correlation. Furthermore, we work out a scaling factor λ as follows:

$$\rho_{implied}^{index} = \rho_{realized}^{index} + \lambda(1 - \rho_{realized}^{index})$$

We can then use it to calculate the implied correlation of each pair of underlying from the historical realized correlation and the scaling factor from index. For example, assume that two stocks A and B are in the index, and the realized correlation is 40%, whereas λ for the index is 15%. The implied correlation for the stock pair is thus:

$$\rho_{implied}^{AB} = 40\% + 15\%(1 - 40\%) = 49\%$$

This is an ad-hoc procedure which is pretty imprecise. Given that it is often difficult to obtain the implied correlation for pair of stocks, basically what is being done here is to make use of some current market data (as given in the calculation of the implied index correlation), and compare the result with the historical data in order to come up with some kind of adjustment.

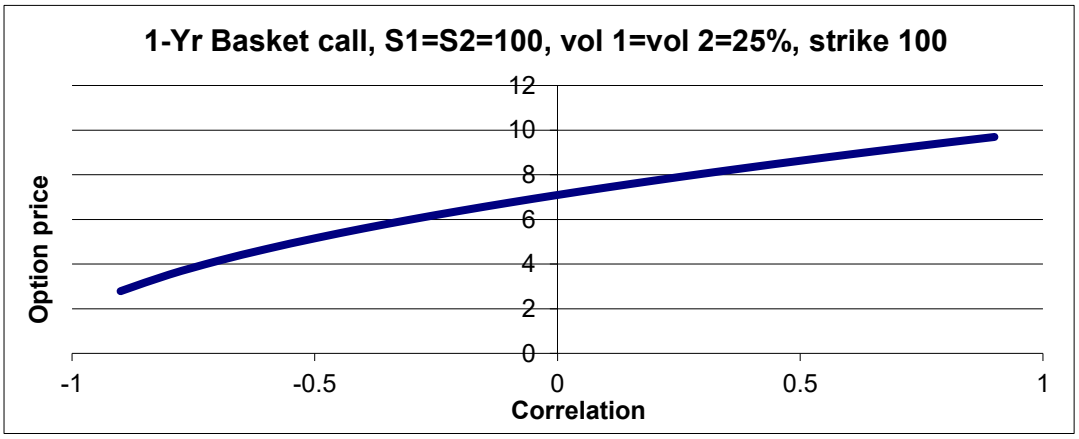
3.C.2.3. Basket call and best-of/worst-of options

The simplest multi-asset option that is dependent on correlation is the basket call

$$Basket(T) = \sum_{i=1}^{n} w_i S_i(T)$$

$$\text{Basket Call} = \max\left[0, Basket(T) - K\right]$$

The basket variance will decrease with decreasing correlation, hence it is a cheaper way to gain exposure to a portfolio of assets. Given that the weights within the basket are fixed, the risk parameters behave similarly as those of the vanilla option. An example of the pricing behavior with two assets:



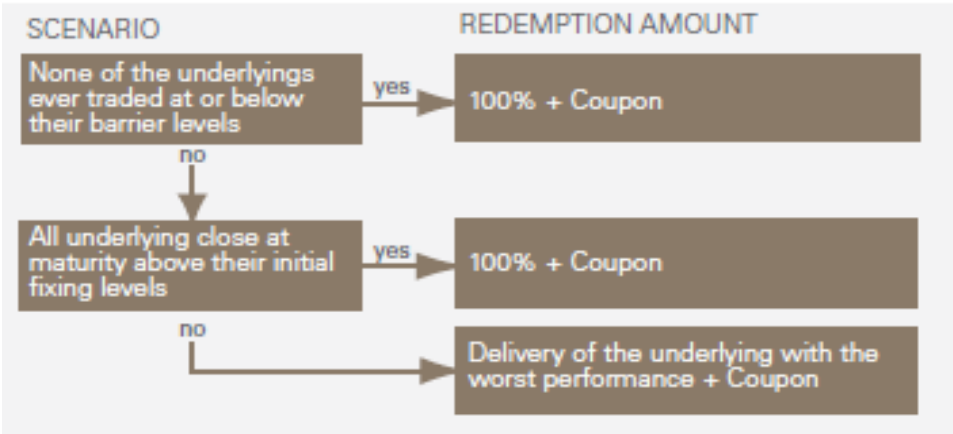
Obviously this is a non-linear relationship. In terms of computation, we can make use of Monte Carlo simulations on simple processes if there are a large number of underlyings, although some products will need to take into account of the volatility skew.

Another popular kind of basic product is the worst-of option:

- Worst-of call: $\max[0, \min[S_1(T), S_2(T), \ldots, S_n(T)] - K]$
- Worst-of-put: $\max[0, K - \min[S_1(T), S_2(T), \ldots, S_n(T)]]$

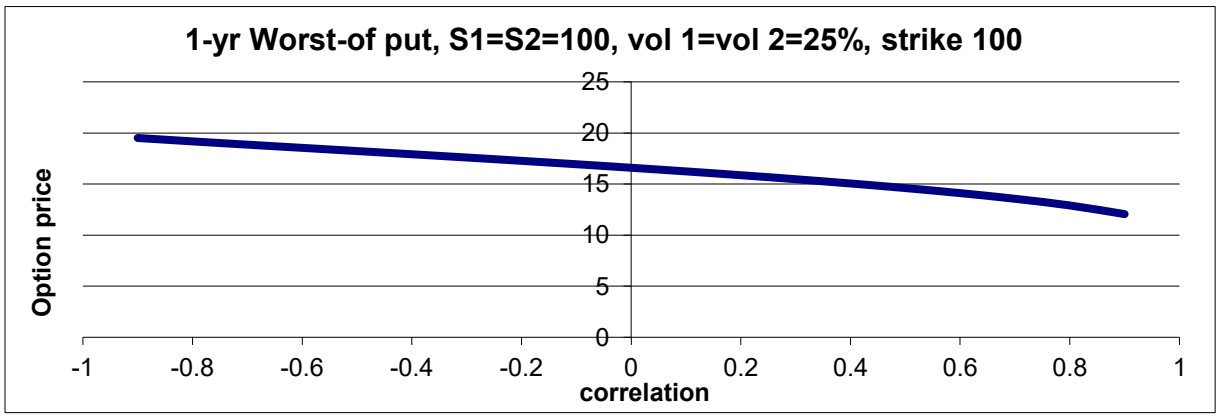
A worst-of call can be much cheaper than a vanilla call, hence it often appears in structured products where investors are buying this option. On the other hand, a worst-of-put is much more expensive than a vanilla put, hence investors often sell these options in order to receive an enhanced yield, especially in equity linked notes. An example is the worst-of reverse convertible note:

Maturity:	1 year
Coupon:	10%
Basket:	Agricultural Bank of China, BYD Company, CNOOC
Barrier:	55% of initial level



The enhanced yield is achievable because the investor of this note is selling a worst-of put option on the three stocks.

We can try to understand the impact of correlation on the pricing with a simple example with only two underlyings:



Unlike the basket call, the worst-of put has an inverse relationship with correlation. As mentioned above, because correlation would often increase in crisis situations, long worst-of put may not perform so well (although volatilities should increase). As investment banks often hold a long position in this instrument, it may suffer a loss if the correlation is not hedged properly (which is difficult to achieve in practice).

For a product with more than two underlying assets, the risks can depend on the relative performance of these assets and can change rapidly. For example, the risk is difficult to manage when the option is near to maturity and at least two assets have similar performances. When one asset outperforms the other, the delta would concentrate on that asset alone, thus it is possible that the delta can switch from one asset to another with small movements in the asset price (see the discussion on Himalaya option in a later section). Similarly, the vega of a particular asset is a function of the positions of the asset with respect to the strike and the relative performances of the assets.

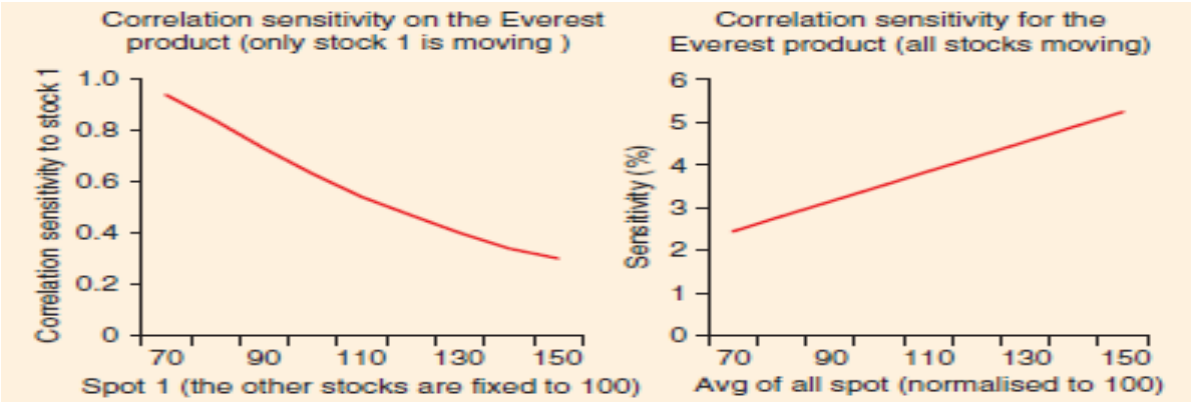
3.C.2.4. The Everest

This is an example of a structure with many underlyings. It belongs to a type of product known as the "mountain range" option, first marketed by Societe Generale in 1998. A typical Everest product has a maturity of longer than 10 years, and with many underlying assets (at least 10 or more). The payoff pattern is simple:

$$\text{Payoff} = \text{Notional} \times [\text{Coupon} + \min(R_i)]$$

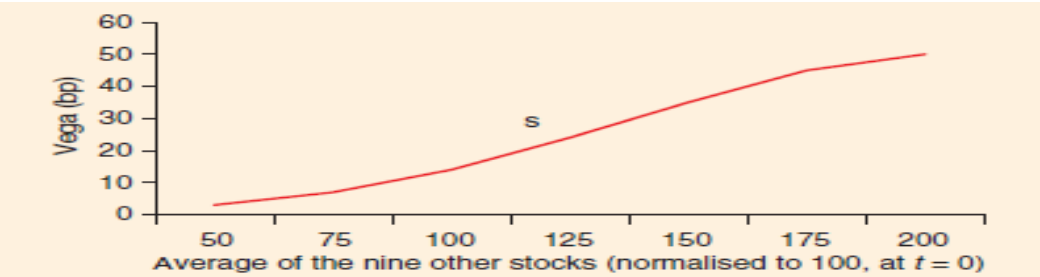
where R_i is the performance of asset i at maturity, which can be negative. If the structure is capital guaranteed, the Coupon cannot be lower than 200%.

The following analysis is taken from Quessette (2002).[3] We look at the impact of correlation to the pricing of the product.



In the left diagram, it is assumed that all the other stocks remain at price = 100, and only stock 1 is moving. When it moves further above this price, the product is less dependent on its price (because it is no longer the stock with the minimum price) and hence the price of this stock is no longer very relevant. As a result, any change in correlation would not have a significant impact on pricing. In the right diagram, all stocks have the same price. We can see that if the price level is higher, the product is more sensitive to correlation, because the absolute price is higher. The whole point of the exercise is to show that it is very difficult to say whether the correlation sensitivity should increase or decrease when the price of stock 1 moves up – it all depends on the prices of the other stocks.

In a similar manner, the vega of a stock is also dependent on the relative performance of the other assets.



Example on the Everest product with a strong vega instability. One stock is fixed at 100, and we move the other stock of the basket and observe the vega of the first

In the diagram above, vega for the stock can increase dramatically even though the stock price has not move, as long as the stock becomes the worst performing asset in the basket. For example, when stock price S_1 = 100 and all other stocks are at 175, vega for S_1 will be high given that it is the worst performing one in the basket. On the other hand, if all other stocks are at 75, the vega of S_1 would be low. Given that the vega position has to be frequently adjusted dynamically throughout the life of the option, one needs to estimate the transaction costs involved in hedging when pricing this product.

[3] Richard Quessette, *New Products, New Risks*, Risk magazine, March 2002.

Reducing pin risk: rainbow payoff

There are many variations for products which involve many underlyings. In the rainbow payoff, unlike the case for a basket option, the weights in the final payoff are allocated to the assets based on their rankings.

For example, there are five underlyings in the product. A weight factor would be assigned initially: w_1, w_2, w_3, w_4, w_5. In calculating the final payoff, we sort the underlying performances in descending order: p_1, p_2, p_3, p_4, p_5. The payoff can then be computed in the usual manner, i.e. payoff: $P = w_1p_1+w_2p_2+w_3p_3+w_4p_4+w_5p_5$.

	Weights				
Worst-of	100%	0%	0%	0%	0%
Smooth 1	90%	10%	0%	0%	0%
Smooth 2	80%	5%	5%	5%	5%
2nd worst	0%	100%	0%	0%	0%
Basket	20%	20%	20%	20%	20%

Compared to the worst-of option, the other structures give less weight to any one single asset, thus the risk would be less concentrated on any underlying.[4]

3.C.2.5. Individually Capped Basket Call (ICBC) and Capped Basket Call (CBC)

3.C.2.5.1. ICBC product description

The characteristics of a product are very much dependent on how it is specified. In many cases, small changes in the specification can lead to very different behavior. We use the CBC and the ICBC to illustrate this point.

A typical product with an Individually Capped Basket Call (ICBC) is defined as follows:

$$R_{i,j} = \frac{S_{i,j}}{S_{0,j}} - 1, CapR_{i,j} = \min[\, R_{i,j}, CAP]$$

$$Coupon(i) = \max\left[0, \frac{1}{N}\sum_{j=1}^{N} CapR_{i,j}\right]$$

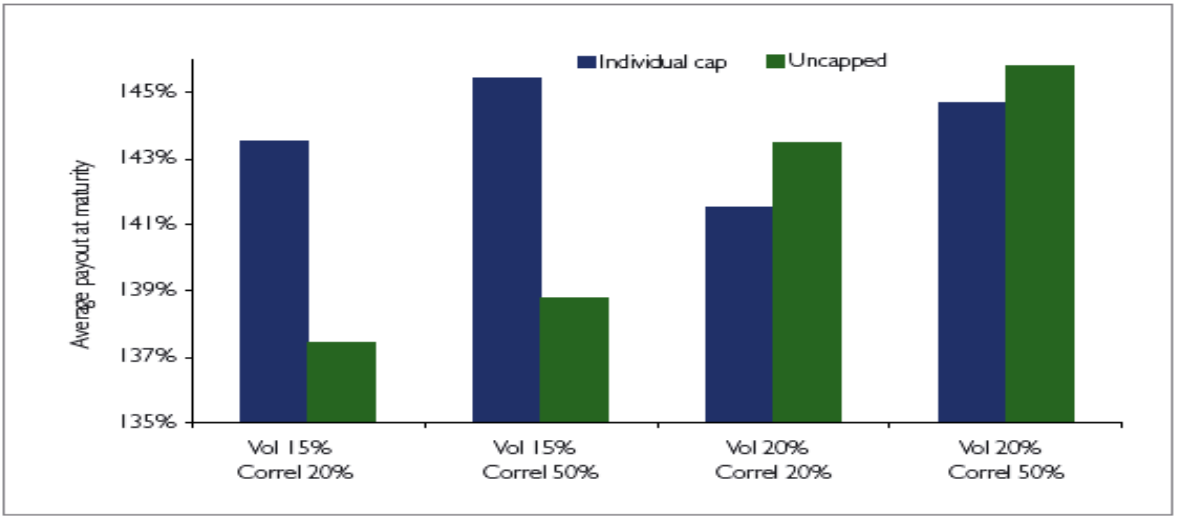
At time i, each asset j's return is capped at the CAP level. Roughly speaking, the investor has a long position in a basket call, but is short a series of calls at the cap strike for each underlying. The investor of this product has a mildly bullish view, hoping that the underlying assets would achieve positive returns, but the returns would be limited. Compared to an unrestricted call, this structure is using the premium from the short call position to increase the participation rate.

ICBC example

Maturity:	5 years
Number of stocks:	5
Capital protection:	100% of initial investment
Final payout:	155% of individual cap basket's growth
Individual cap:	175% of initial asset's value

[4] C.f. Hans Buehler, www.quantitative-research.de.

A simulation is performed with the following parameters: volatilities 15% and 20%, average correlation 20% and 50%, interest rate = 5%.



Source: Citigroup

The ICBC structure outperforms the basket call at low volatilities mainly because of the increased participation. When the volatility of the basket is at 15% and the correlation is 20%, there is relatively little chance for each stock to hit the cap level, and hence the return is high because of the big multiplier (155%). This argument still applies when correlation moves higher. However, when the stock volatility reaches 20%, it is then more likely for some of the stocks to hit the cap, where the payoff is limited. In general, it is not possible to make a general conclusion as to when the ICBC will have a higher payout than a non-capped call, because the result will be based on a combination of many factors.

3.C.2.5.2. Comparison with the CBC

We can compare the characteristics of the ICBC with a capped basket call (CBC), where the payoff is

$$CBC = \max\left[0, \min\left(\frac{1}{N}\sum_{j=1}^{N} R_{i,j}, CAP\right)\right]$$

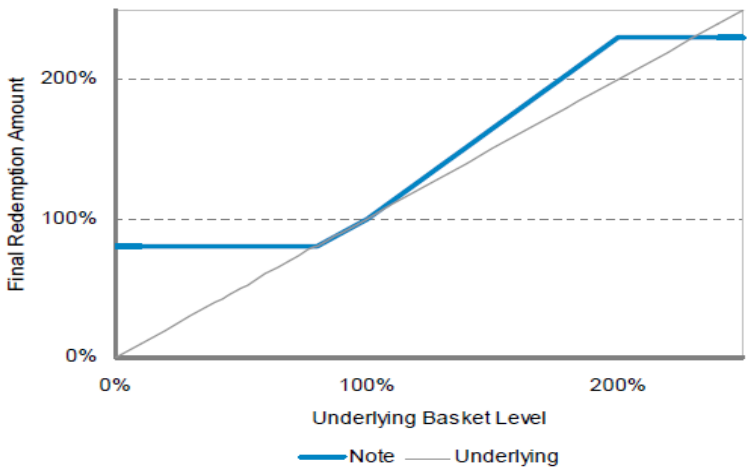
Basically, the CBC is a call spread on the basket. This option can be much more expensive than the ICBC, because in the case of the ICBC, one stock which performs poorly cannot be compensated by the strong positive performance of another (because the upside is capped), whereas for CBC, it is the average performance that counts. For example, an ICBC with 2 stocks is compared against a CBC. The strike is 100, and the individual cap is set to 130 for the ICBC and also to 130 for the CBC. If the final stock prices are 110 and 150, the ICBC's payoff is only [min(110,130)+min(150,130])/2 – 100] = 20, whereas the CBC's payoff is min[(110+150)/2,130] – 100 = 30.

A real example is a product known as an Individually Capped Note on China Urbanization Winners, issued by Morgan Stanley in April 2010. The note has a maturity of 5 years and is denominated in USD. It is linked to a basket of 8 stocks: China Construction Bank, China Life Insurance, China Merchants Bank, China Overseas Land, China Unicom, Industrial and Commercial Bank of China, Maanshan Iron & Steel, and Tencent Holdings. The final redemption value is based on the formula:

Final redemption = max[100% + participation rate x average component performance, 80%]

Each basket component's performance is capped to 100%. The participation rate is determined as follows:

- If the final basket performance exceeds the initial reference price, participation rate is 130%
- If the final basket performance is below the initial reference price, participation rate is 100%



Underlying Basket Level	Payout
70%	80%
80%	80%
100%	100%
150%	165%
200%	230%
220%	230%

Source: Morgan Stanley

The attractive features in this product include:

- Downside protection, with the loss limited to 20%
- Enhanced participation in the upside (between 100% and 200% of initial level)
- While the maximum payout is limited, the cap is high enough (230%) to give a satisfactory return

The important consideration for a structurer is to be able to design this product so that the pricing is reasonable. It is a matter of selecting the stocks in the basket, deciding on the participation, the cap level, and the downside protection level.

3.C.2.5.3. ICBC and CBC: volatility and correlation sensitivity

ICBC is a short dispersion product, i.e. when volatility is high and the correlation is low, the downside effect is more prominent than the upside effect (because the upside is capped, as explained above), making the product cheaper. As a simple example, let the cap be set to 120, strike is 100 and the current price is 100 for each stock. If the volatility is low and the correlation is low, one stock can move to 95 and the other stock price is 115, the payout is 5. However, when the volatility is higher, the stock prices may move to 85 and 125, but since a cap is imposed, the payout is now only (85+120)/2 – 100 = 2.5.

On the other hand, the CBC is like a normal call spread, hence higher volatility would make it more expensive when it is at-the-money. It is also a long correlation position, as higher correlation leads to a higher basket variance. The vega will only become negative when it is near to the higher strike (as in the case of a normal call spread).

3.C.2.6. Himalaya option

3.C.2.6.1. Product description

We look at another structure in the "mountain range" product called the Himalaya option. It is linked to the performance of a basket of N assets. In the basic form, it is a call option on the average of the maximum performance of b assets ($b<N$), with the best performing assets being removed in regular periods during the term of the product. The rationale behind the product is that shares can be chosen from different sectors, so as to capture the best performance of each stock during the business cycles. Some of these structures can even offer leveraged returns (through a high participation rate).

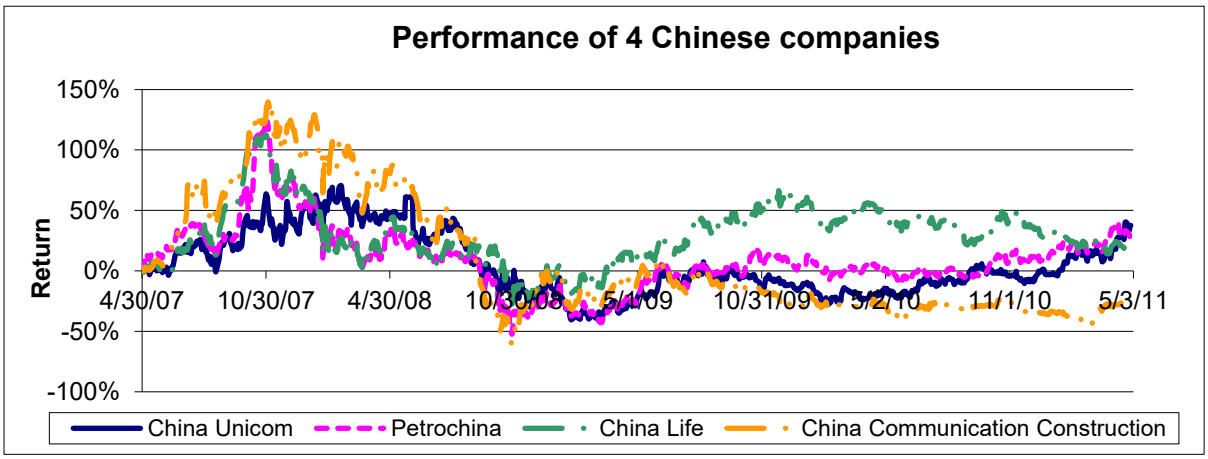
An example is given below:

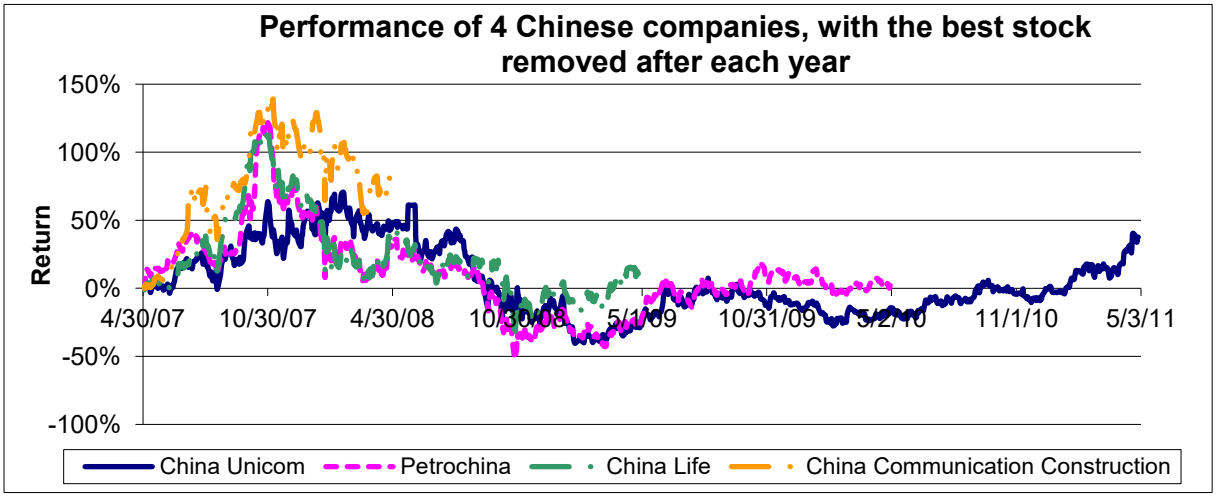
Underlying basket:	China Unicom, Petrochina, China Life Insurance, China Communications Construction
Maturity:	4 years
Strike (K):	100%

Coupon = $$120\% \times \max\left[0, \frac{1}{n} Perf(T) - K\right]$$

$$Perf(t_i) = Perf(t_{i-1}) + \max \frac{S_{t_i}^j}{S_0^j}, j = 1 \text{ to } 4$$

The best performing stock is removed each year





	Cumulative returns (%)			
	762	**857**	**2628**	**1800**
Yr 1	47.0	30.2	38.2	***80.5***
Yr 2	-22.7	-23.0	***11.9***	-9.0
Yr 3	-14.5	***2.1***	46.2	-27.5
Yr 4	***37.6***	25.3	12.9	-30.1

The Himalaya option coupon is thus: 1.2x(80.5+11.9+2.1+37.6)/400 = 39.63%, with the annualized return = (1.3963)^(1/4) – 1 = 8.70%.

This is a popular structure, as it looks like a best-of product with leverage. However, this option is much cheaper than the best-of option, as the worst performing asset can have a negative return and a (relatively) long maturity compared to the other assets, hence dragging down the overall performance. In the previous example, the option benefits from the huge outperformance of China Communications Construction in the first year.

3.C.2.6.2. Risk analysis

The discussion and the following diagrams are mostly taken from Overhaus (2002).[5]
There are a few variations of this structure. The following types are common:

$$(3): M\sum_{i=1}^{b}\left(\frac{S_{n(i)}(T_i)}{S_{n(i)}(T_0)}-1\right), b<N$$

Standard type: Only pay the best b assets out of N (no floor) (i.e. number of payout periods < N)

$$(4): M\sum_{i=1}^{N}\max\left\{0,\left(\frac{S_{n(i)}(T_i)}{S_{n(i)}(T_0)}-1\right)\right\}$$

Floor each asset's performance to 0

$$(5): M\max\left\{0,\sum_{i=1}^{N}\left(\frac{S_{n(i)}(T_i)}{S_{n(i)}(T_0)}-1\right)\right\}$$

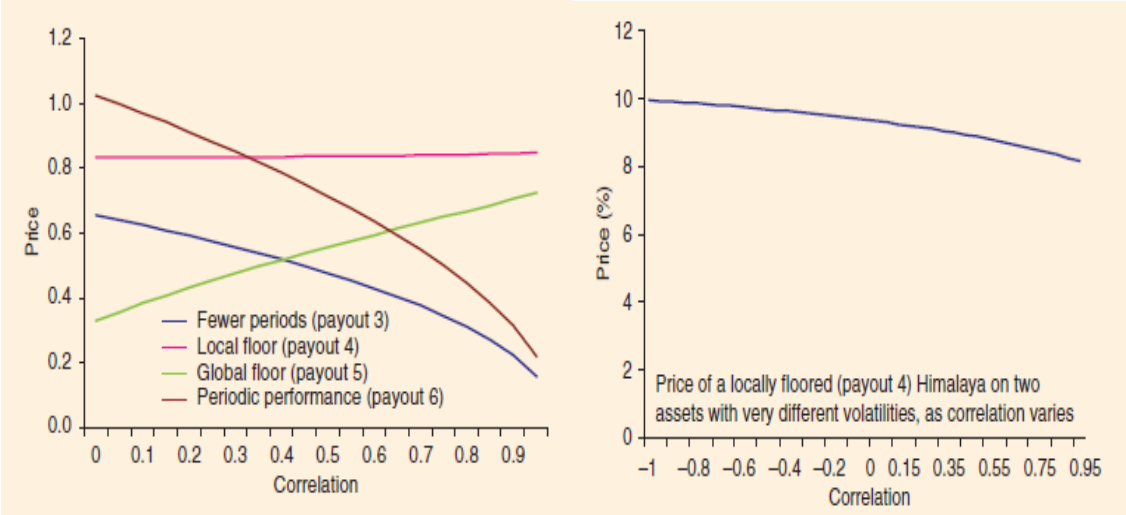
Global floor to the coupon, i.e. capital guaranteed

$$(6): M\sum_{i=1}^{N}\left(\frac{S_{n(i)}(T_i)}{S_{n(i)}(T_{i-1})}-1\right)$$

Paying the best performance over the current period (no floor)

While the basic theme of these structures is similar, their behaviors can be very different. Firstly, we can look at the sensitivity to changes in correlations among assets.

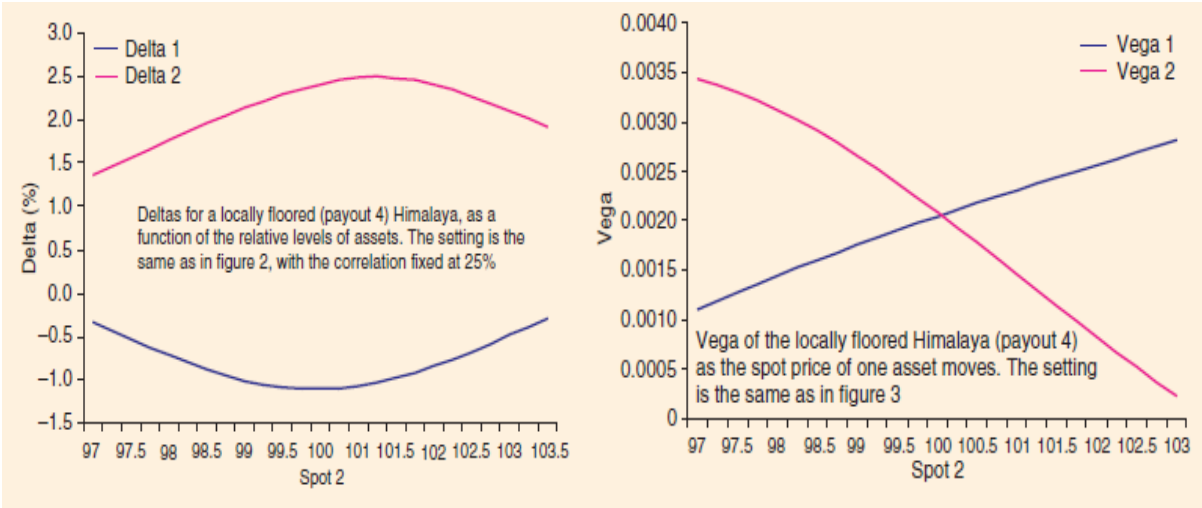
[5] Marcus Overhaus, "Himalaya Options," *Risk Magazine*, March 2002.



The graph on the left assumes that all assets have prices = 100 and volatility = 20%, with a constant correlation among all assets, N=6 assets, and there are 6 periods for all payouts except for payout 3, which has 4 periods:

- Payout 3 is short correlation – it is close to a best-of option (as discussed above; see section xxx). Note that the correlation sensitivity will become less if the number of periods b approaches N.
- Payout 5 is long correlation, because it is like a basket call option; increase correlation will increase the variance of the basket, leading to a higher price.
- Payout 6 is strongly short correlation, as this is similar to a sum of best-of options, where each component is short correlation.
- The analysis of payout 4 can be more complex. In the left graph, it is not too sensitive to correlation and shows an almost horizontal profile. This is only true because all options are ATM. In the graph on the right, it is assumed that there are two assets, volatility of S_1 = 50%, volatility of S_2 = 20%, and just before the first observation date T_1, $S_1(T_{1-})= S_1(T_0)$, $S_2(T_{1-})$=0.97x $S_2(T_0)$. At T_1, after the removal of the best performing stock (i.e. the stock with a high price), the position becomes a near-the-money option on either S_1 or S_2, depending on which has performed worse at T_1. Assume that the option for S_1 will be worth 20% and that of S_2 will be worth 8% (because of different volatilities). In this scenario, if correlation is close to 100%, it is unlikely that S_2 can outperform S_1, hence the value is low, but if correlation is –100%, it is more likely that S_2 can outperform S_1, hence increasing the option value. Thus the correlation sensitivity is negative. However, the correlation sensitivity is reversed if S_2 has slightly outperformed S_1 at T_{1-}, i.e. the higher the correlation, the higher the price.

The sensitivity of delta and vega are equally complex. Using similar parameters as before, i.e. volatility of S_1 = 50%, volatility of S_2 = 20%, correlation = 25%, and just before the first observation date, $S_1(T_{1-})= S_1(T_0)$=100 (i.e. return = 0%).

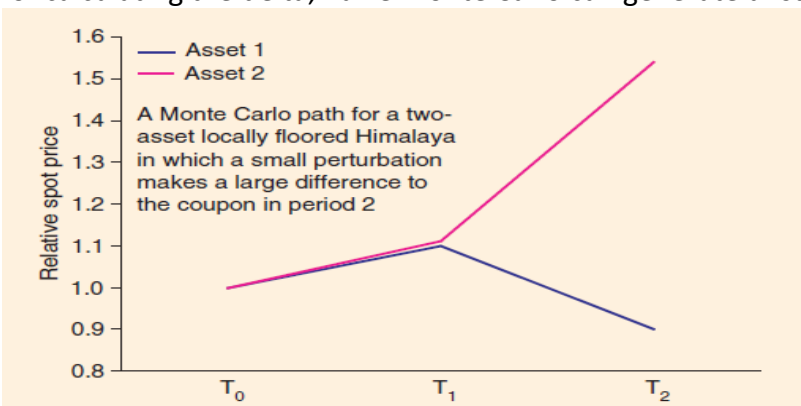


Starting from a level which is below the price of S_1 (=100), when the price of S_2 goes up but below 100, it is still likely that S_1 will be the better performing asset, but the 1-year option on S_2 is more in-the-money, causing its delta to go up. However, when S_2 goes up further, it becomes the better performing asset, and hence its delta will be less as it would replace S_1 as the better performing asset at time T_1, which means that there is less chance for the option to be needed after time T_1. The reverse argument will apply to S_1 as its delta is just a mirror image of that of S_2.

The vega profile can be explained in a similar manner. As the price of S_2 moves from 97 to 103, the likelihood that it will become the better performing asset increases (and thus will be "knocked out"), hence its vega will decrease, and the vega is shifted from one asset to the other.

For each asset, a dynamic allocation of vega across different time buckets is required, and it depends on the probability of survival into the corresponding period. If one asset outperforms all the others, the vega in all time buckets will drop considerably (because it will be knock out in the next valuation date), whereas the vegas for the other assets in the first bucket will drop, but the other assets' vegas in the forward periods will rise.

In terms of calculating the delta, naïve Monte Carlo can generate unstable results.



In the figure above, asset 2 performs better in period 1, but if there is a small perturbation so that asset 1 is deemed to perform better instead (and the level is fixed), the final payout can be dramatically different. It means that the payout can be very sensitive to small relative movements in the asset prices, especially when their returns at the observation dates are close to each other.

3.C.2.7. Altiplano Notes

3.C.2.7.1. Product description

Another example of a mountain range product is the Altiplano notes, which is also a structured product linked to a basket of N assets. It entitles the holder to receive a large coupon (e.g. 30%) if none of the assets in the basket has fallen below a pre-determined barrier during a given time period. Otherwise, a different payoff is calculated at maturity, e.g. based on a vanilla call option on a basket. There can be many variations, such as:

- The vanilla call can be replaced by other types of options, e.g. Asian call
- A variable coupon is paid depending on how many assets hit the barrier

A typical example is as follows:

Maturity:	3 years
Initial strike:	100%
Payoff:	Coupon (if barrier is not hit) or a vanilla call on a basket (strike is reset every year)
Coupon:	10%
Barrier:	90%
Basket:	9 blue-chip stocks UBS, Deutsche Bank, BNP Paribas, AIG, General Electric, Wal-Mart, Citigroup, Goldman Sachs, Bank of America

Another structure was issued by Deutsche Bank on February 28, 2011:

Maturity:	5 years
Coupon:	4% every 6 months, if each stock in the basket is above the barrier; 0% otherwise
Barrier:	100% of initial level
Basket:	Bayer AG, GlaxoSmithKline, Novartis, Pfizer, Sanofi-Aventis

In this example, the leverage is lower given the smaller coupon.

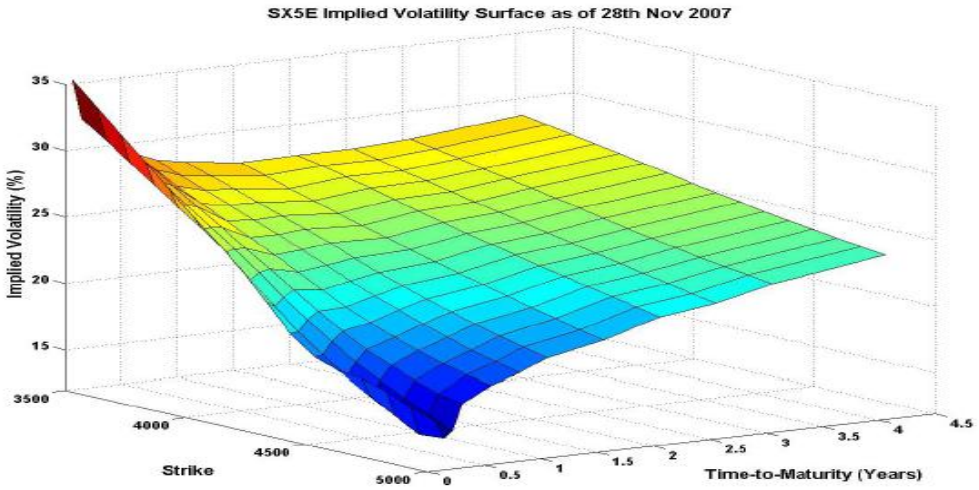
3.C.2.7.2. Risk analysis

The main risk of this product is a digital risk: a big coupon is paid depending on whether one of the stocks will (not) hit the barrier. It is sensitive to the correlation of the assets in the basket, as well as the volatility skew (given that there are barriers). Effectively the huge digital risk will concentrate on one stock (i.e. the worst performing), which means that the risks will not be spread among the different stocks in the basket. For prudent risk management, the issuer needs to limit the notional size of the product, taking into account the liquidity of the names within the basket.

As in the case of products with many underlyings, correlation between the assets is one of the significant pricing factors. In many cases, the underlyings selected for a product come from a customized basket, with a particular research theme. Hedging the correlation can only be performed in an approximate manner. For example, it may be possible to create some kind of proxy hedge. For long correlation, the investment bank can go long on straddles on an index and short straddles on each of the individual components (but need to delta hedge each position in order to create an exposure on correlation only, which is tedious). An alternative is to use variance swaps to replicate the above positions. In fact investment banks are often short correlation (through the structured products), but some risks have been transferred to hedge funds which are willing to take positions on these risks.

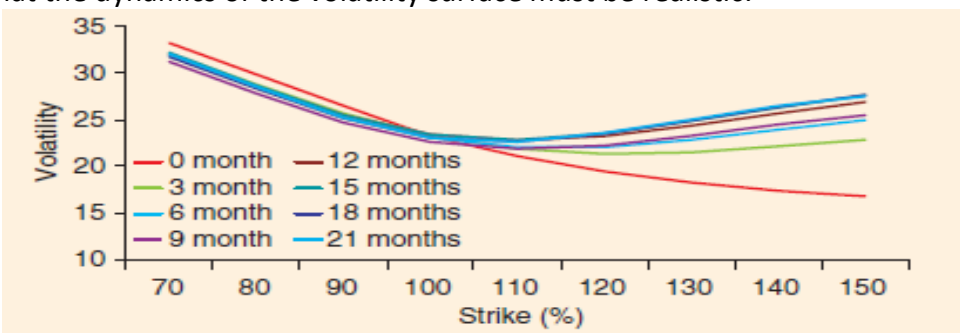
3.C.2.8. Structures that are dependent on forward volatility skew

It is well known that vanilla options such as call spreads and digital options are dependent on the volatility skew. In addition, many exotic equity structures incorporate forward starting options, and are thus exposed to the risk of the forward volatility skew. While different models may be able to fit the current market prices of vanilla options, the possible forward volatility skew allowed under each model can be very different.



The diagram above shows an implied volatility surface of Eurostoxx 50 as of Nov 28, 2007.[6] We can see that at each maturity, the skew is mostly downward sloping, i.e. option at a higher strike will have a smaller volatility, although the skew is not so significant if the maturity is longer.

While it may be possible to fit all the vanilla option prices in the market, it does not mean that the dynamics of the volatility surface must be realistic.



The above diagram shows an example of the possible 3-month forward skew for Eurostoxx 50 generated by a local volatility model (Quessette, 2002). It is seen that the model generates some kind of "smile" instead of skew for longer maturities. This shows the weakness of the local volatility model: even after 3 months, the skew is very different from what is expected from the market. Hence if we need to price a product which is stronger dependent on the forward volatility skew, local volatility model is not a good choice.

Risk management of forward starting options

Additional risk parameters are needed for the management of exotic structures. Two useful parameters are volga and vanna:

[6] Source: Martin Haugh, Quantitative Risk Management class notes, Columbia University, Spring 2010.

- Volga (or vega gamma) – change in vega when volatility moves. A vega convex payoff means that when volatility goes up, vega increases.
- Vanna – sensitivity of the delta to a change in volatility, or the sensitivity of the vega to a change in underlying spot

These parameters often affect exotic structures that are linked to forward starting options, e.g. cliquet products. Also refer to the example of the Himalaya structure in section xxx.

3.C.2.9. Locally capped, globally floored (LCGF) cliquet and the Napoleon structure

An LCGF cliquet is sometimes known as the "Predator." At each date, the coupon is based on the short term performance of each period, subject to a cap p_i:

$$\text{coupon} = \min\left[\frac{S(T_{i+1})}{S(T_i)} - 1, p_i\right]$$

At maturity, there is a minimum guaranteed total coupon c:

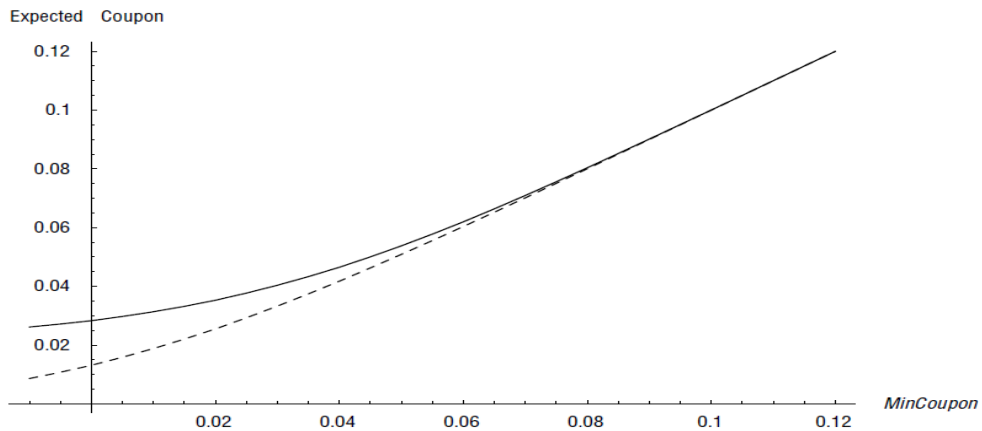
$$\text{final coupon} = \max\left\{\sum \min\left[\frac{S(T_{i+1})}{S(T_i)} - 1, p_i\right], c\right\}$$

This resembles the payoff of the ICBC structure, where the returns are taken through time instead of over a basket of assets.

An LCGF cliquet was issued by Mediobanca S.p.A. on December 2, 2002, with the following terms:

Maturity:	3 years
Underlying:	Eurostoxx 50 index
Coupon:	$\max\left\{\left[\sum_{t=1}^{12} \min(\max(r_t, -1\%), 1\%)\right], 2\%\right\}, r_t = \frac{S_t}{S_{t-1}} - 1$, paid annually

Basically, each monthly coupon has a range of –1% to 1%, with the annual coupon being capped at 12% and floored at 2%. We can immediately see that if not for the global floor, the structure is just a strip of forward call spreads, and the pricing should not be difficult.

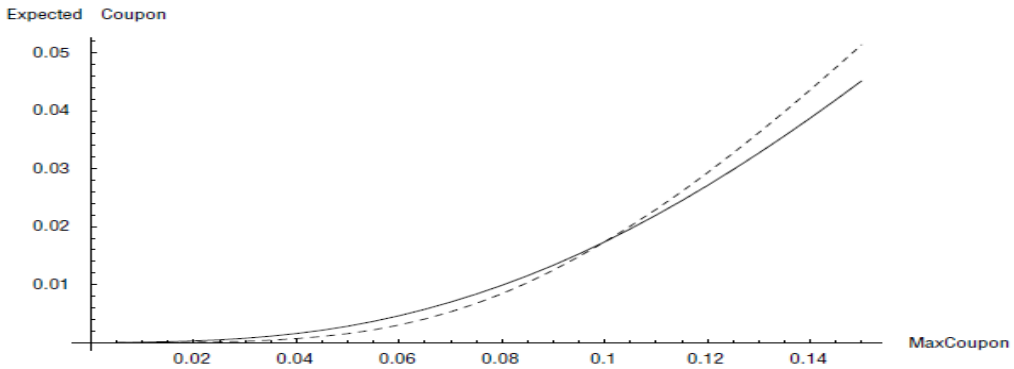


The solid line shows the pricing of the LCGF cliquet under a Heston stochastic volatility model, while the dashed line is the prices from a local volatility model (Gatheral 2006).[7] Since the local volatility model will generate a forward skew which is too flat, it will substantially underprice the deal, as seen in the graph above.

An example of a Napoleon structure is a product issued by Mediobanca on December 20, 2002, which is a capital guaranteed structure"

Maturity: 3 years
Coupon: max[0, MaxCoupon + R_i], paid annually
MaxCoupon = 10%; R_i is the average of the worst (possibly negative) monthly returns of S&P500, Eurostoxx 50, and Nikkei 225

Given that the payoff is similar to that of shorting a downside put, it is expected that the stochastic volatility model (with more realistic downside skew) would give a lower price than the local volatility model.



It turns out that depending on the MaxCoupon, the price given by local volatility model can be higher or lower than the stochastic volatility version (c.f. Gatheral 2006:127-9). This is a surprising result which is not well understood. In fact, there was serious mis-pricing of this product in 2002, when this product was heavily traded by many banks in Europe. To avoid the pricing error, a correct modeling of vega convexity is required, especially in volatile markets.

3.B.2.10. Final remarks

After the financial crisis from 2008, there has been less demand for complex structured products. Both the investor and the investment have suffered losses. The lesson learnt is that one needs to fully understand the payoff pattern and the associated risks. In particular, second order risk parameters can be very important

For the banks offering the products, a careful analysis is required in deciding what kinds of features should be included in the pricing models. Implementing the most sophisticated model is not always the best solution as there is a cost in terms of efficiency and parameters requirement. Furthermore, there is often a need to think of creative ways to hedge the many different kinds of risks that are prevalent for exotic structures.

[7] Jim Gatheral, *The Volatility Surface: A Practitioner's Guide*, Wiley, 2006, p.124